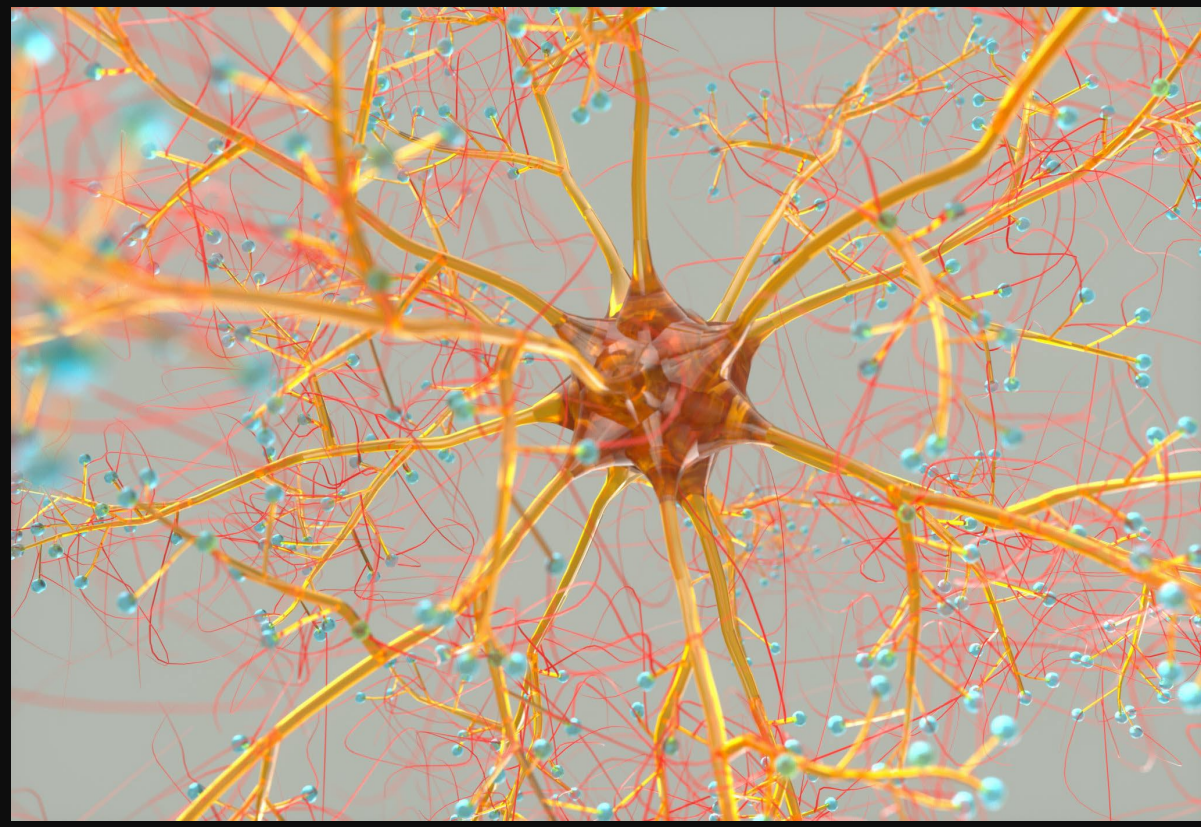
Neurology of the Upper Limb

Prof Isaac Sorinola

Learning Outcomes

- Give an account of the distribution of cutaneous nerves of the upper limb
- Describe the pattern of dermatome innervation of the upper limb
- Describe the myotome map of the upper limb, and list the nerves responsible for movement around each joint of the upper limb
- List the origin, course and distribution of the brachial plexus
- Identify the major sites of injury to the nerves of the upper limb and outline common causes and consequences of damage to them



Quick recap from Neurology of the lower limb



Define the term 'dermatome' using an example.

Nobody has responded yet.

Hang tight! Responses are coming in.

Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

Using an example, define the term 'myotome'.

Nobody has responded yet.

Hang tight! Responses are coming in.

Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

Nerves entering the lower limb carry fibres from all of the following except

- L1
- T12
- L2
- L3
- All of the above

Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

The sacral plexus is formed by anterior rami from L4, L5, S1, S2, S3 and S4



Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

Learning Outcomes

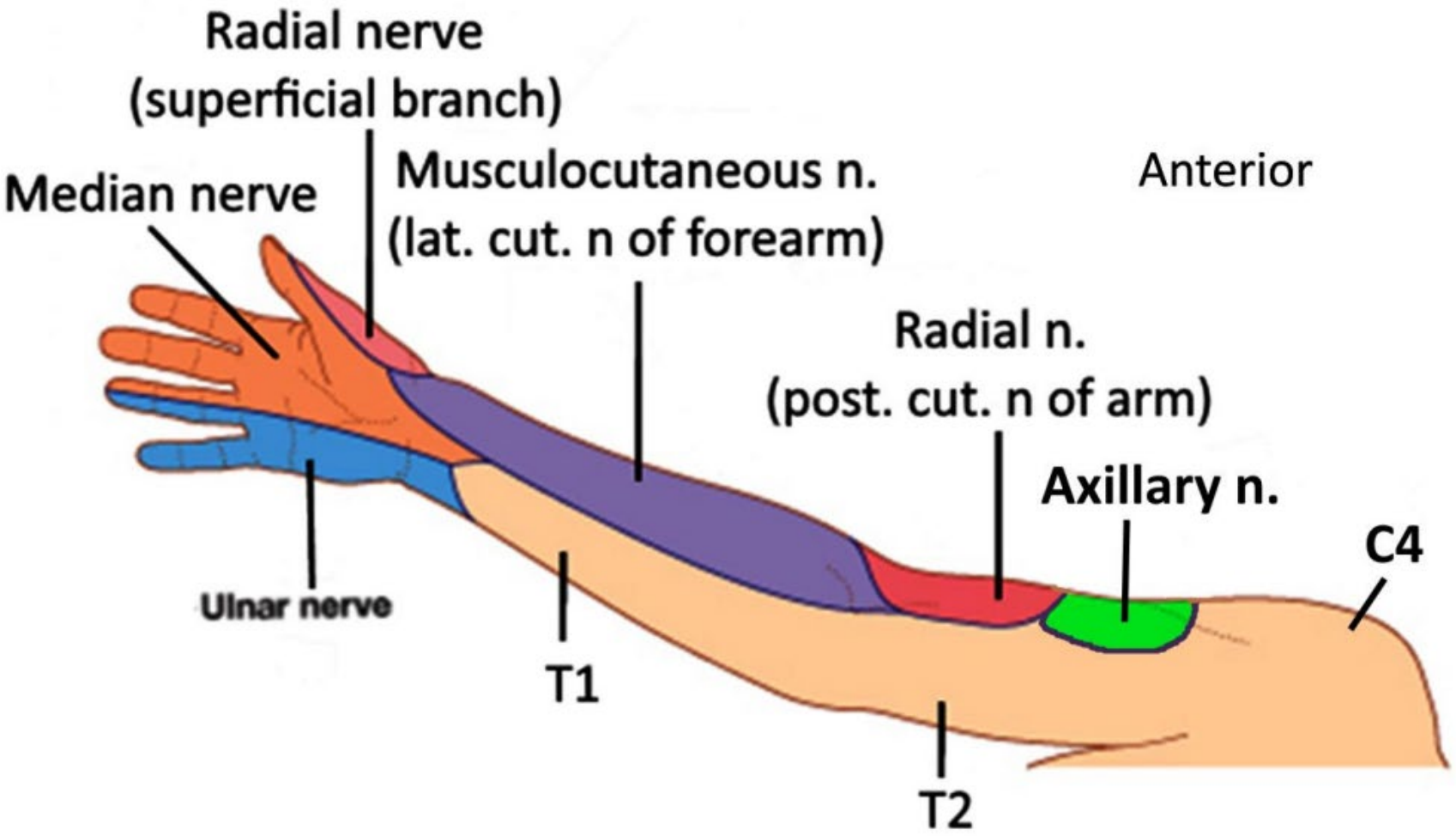
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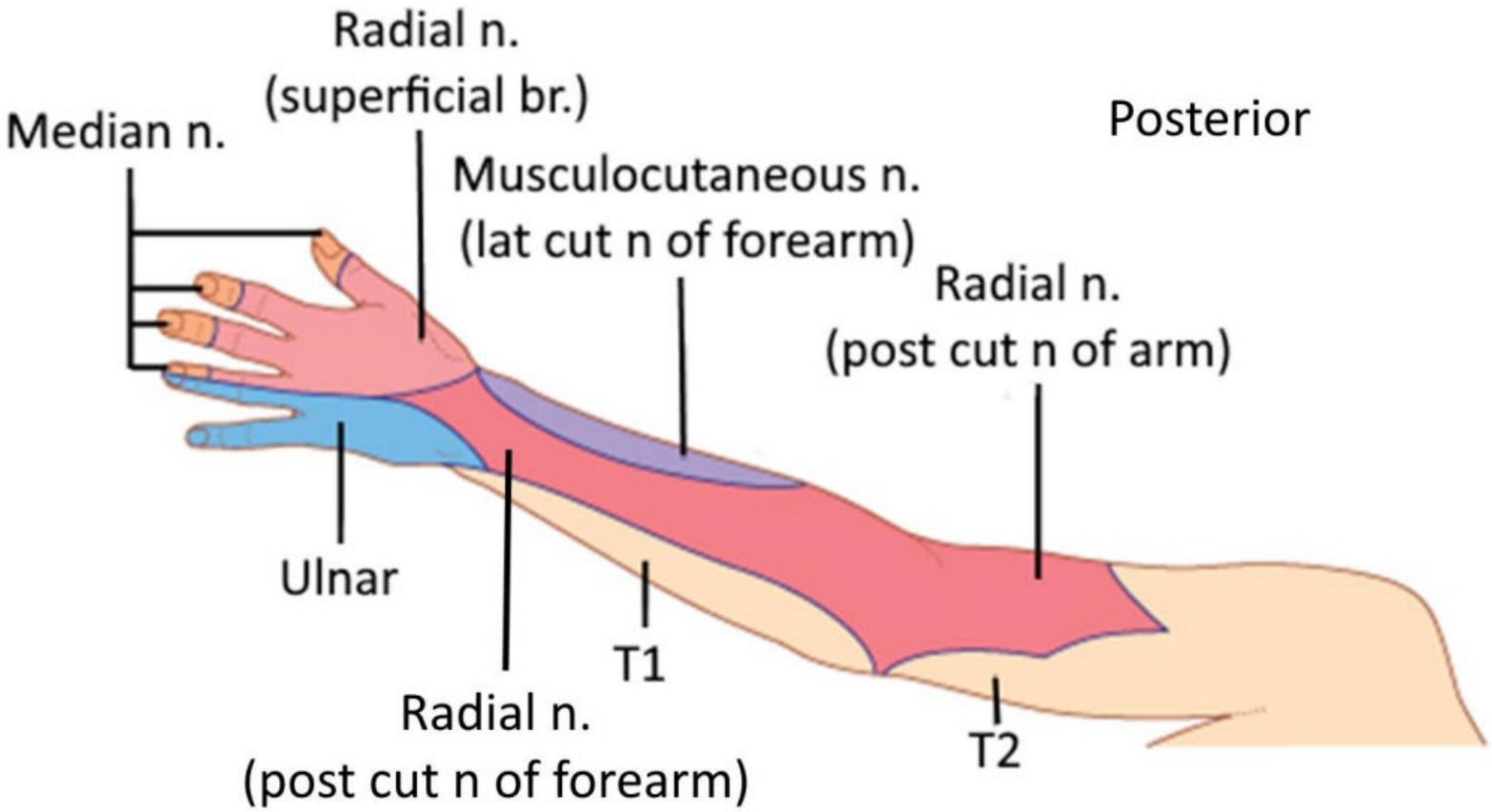
Upper Limb Nerve Injuries

- What can we test?
  - Spinal roots
    - Dermatomes
    - Myotomes
  - Peripheral Nerves
    - Cutaneous branches
    - Motor branches

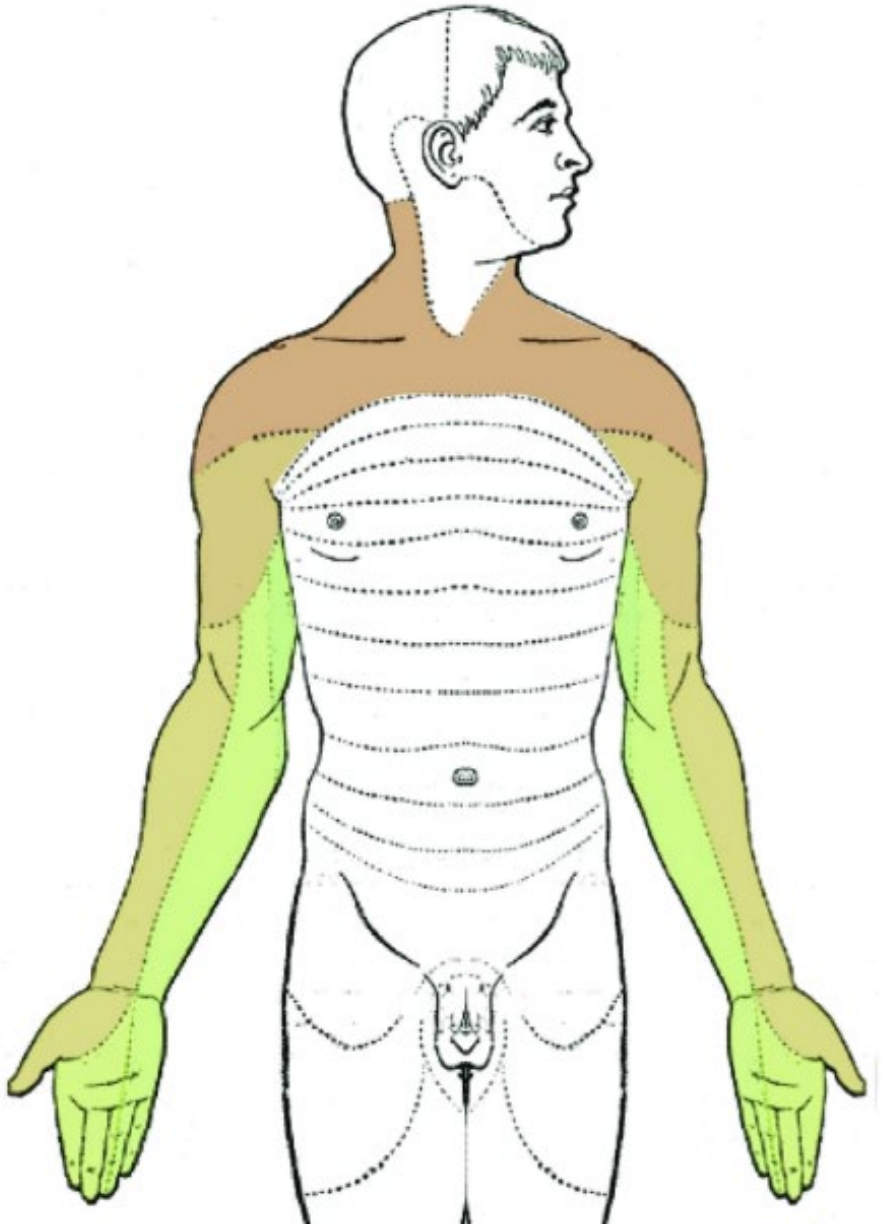
Cutaneous Nerve Map (Anterior View)



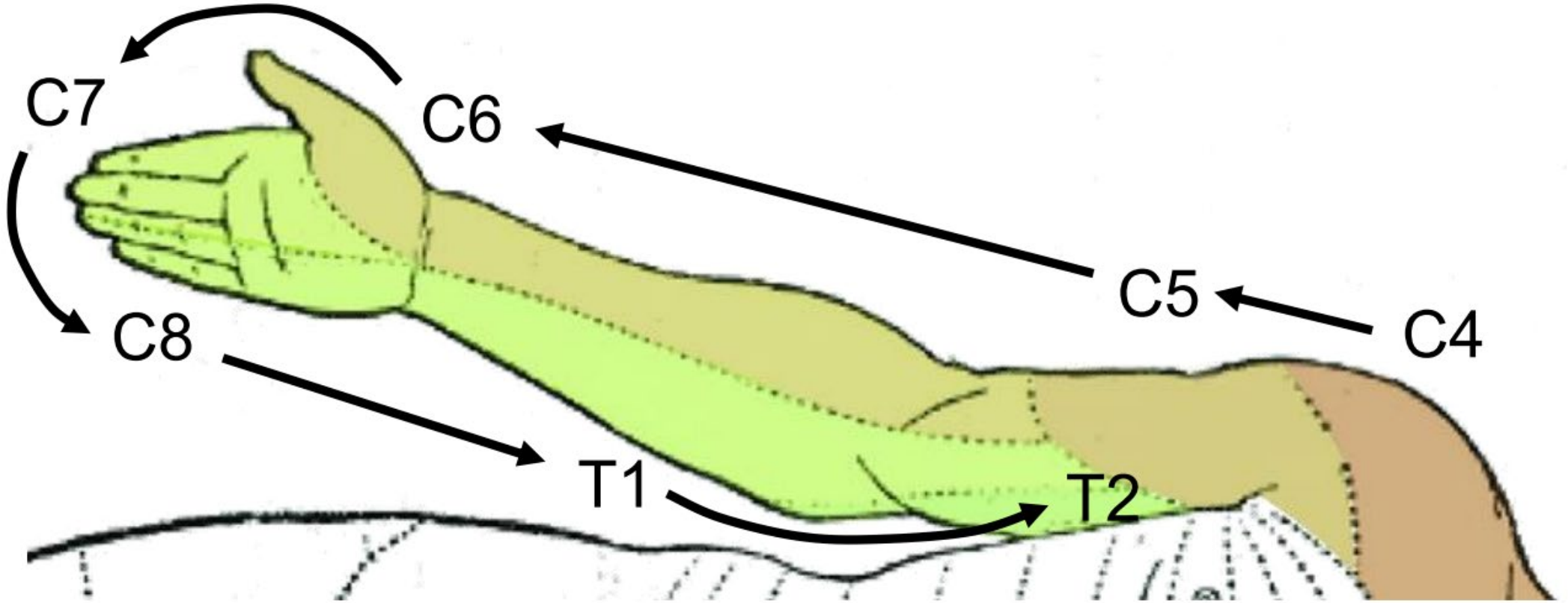
Cutaneous Nerve Map (Posterior View)



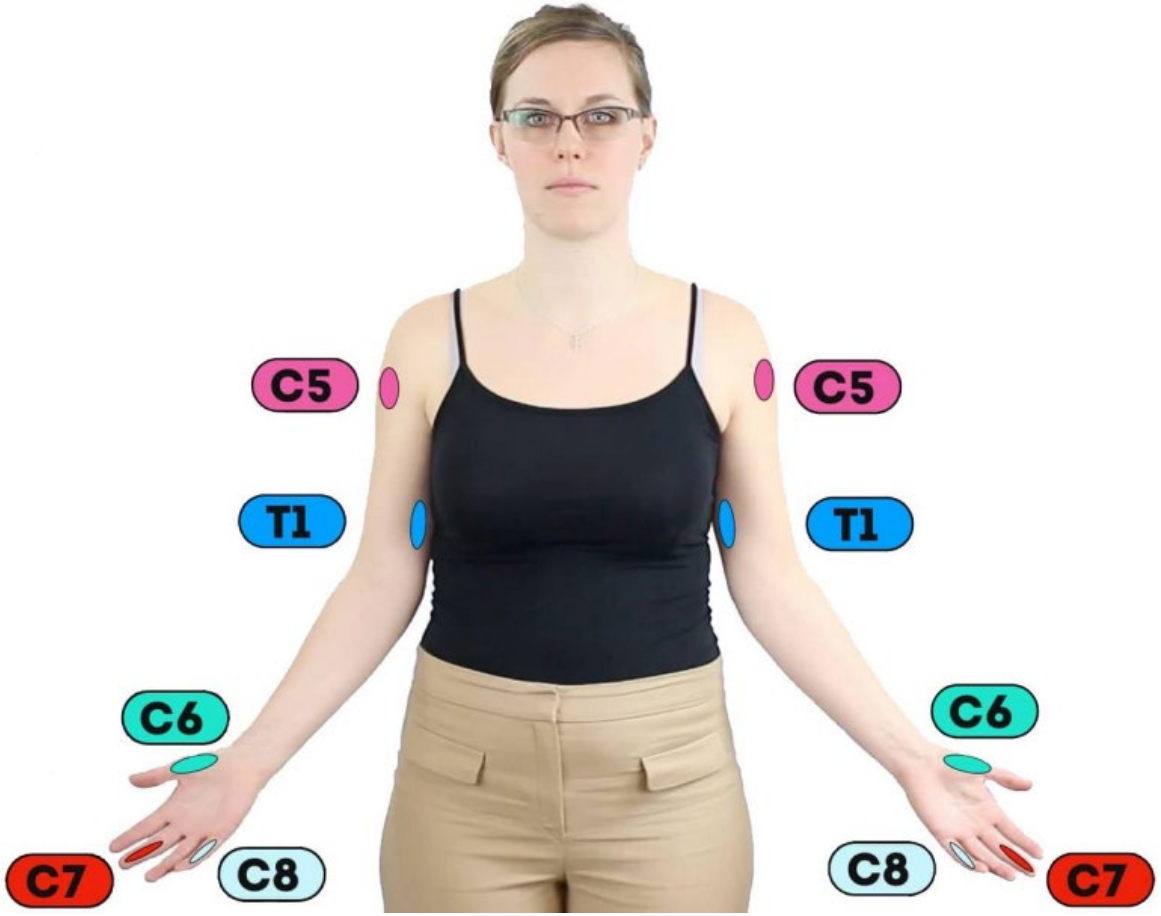
Dermatomes of the Upper Limb



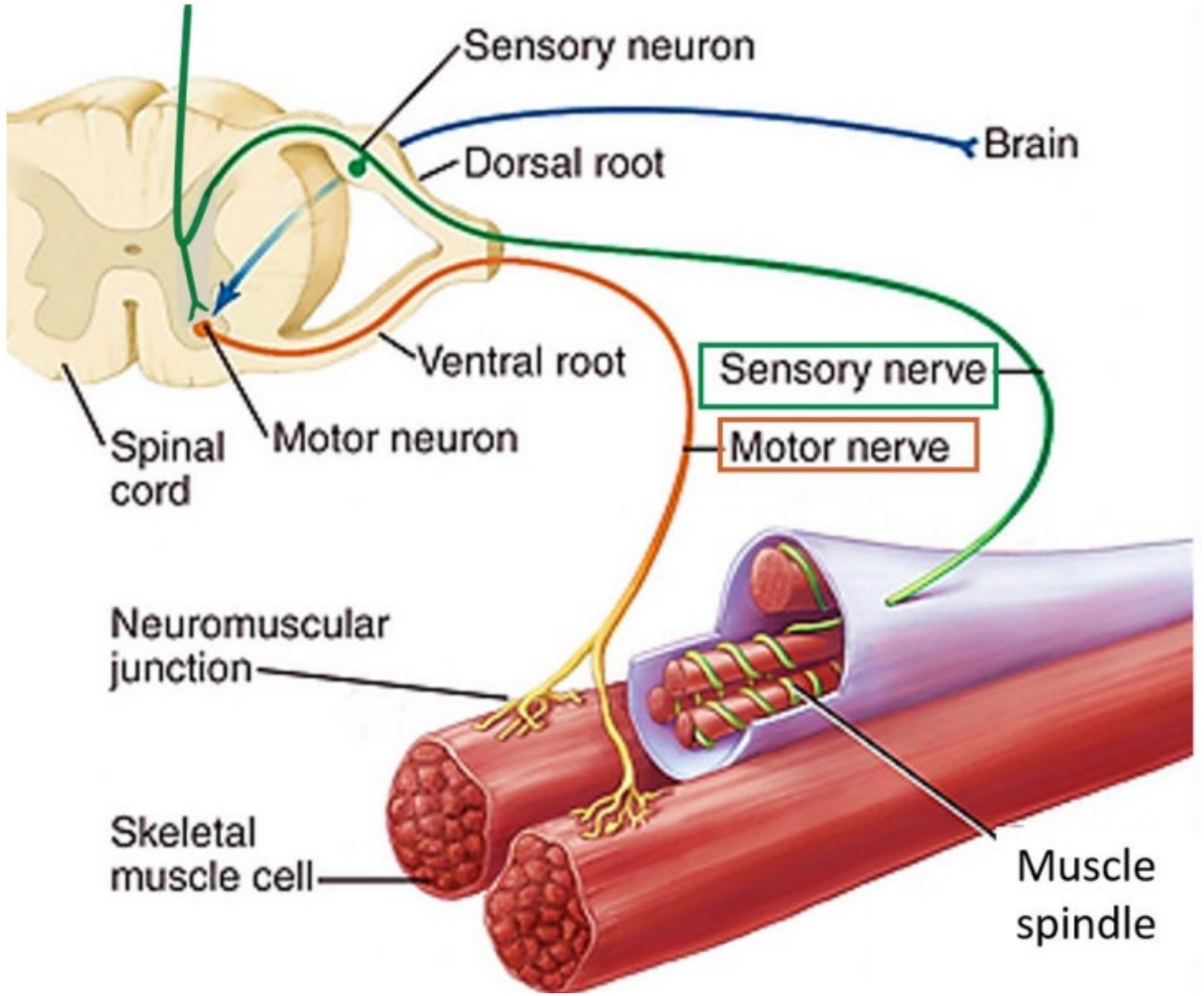
Dermatome of the Upper Limb



Dermatome Testing Points



Motor Nerves



Myotomes of the Upper Limb

Joint	Movement	Segmental value
Shoulder	Abduction, lateral rotation	C5
	Adduction, medial rotation	C6, 7, 8
Elbow	Flexion	C5, 6
	Extension	C7, 8
Forearm	Pronation	C7, 8
	Supination	C6
Wrist	Flexion	C6, 7
	Extension	C6, 7
Fingers	Flexion, Extension (long muscles)	C7, 8
Hand	Intrinsic muscles	C8, TI

C5 Roots Test





https://youtu.be/a4_RPLsaoRM

C6 Root Test







C7 Root Test





C7 Root Test



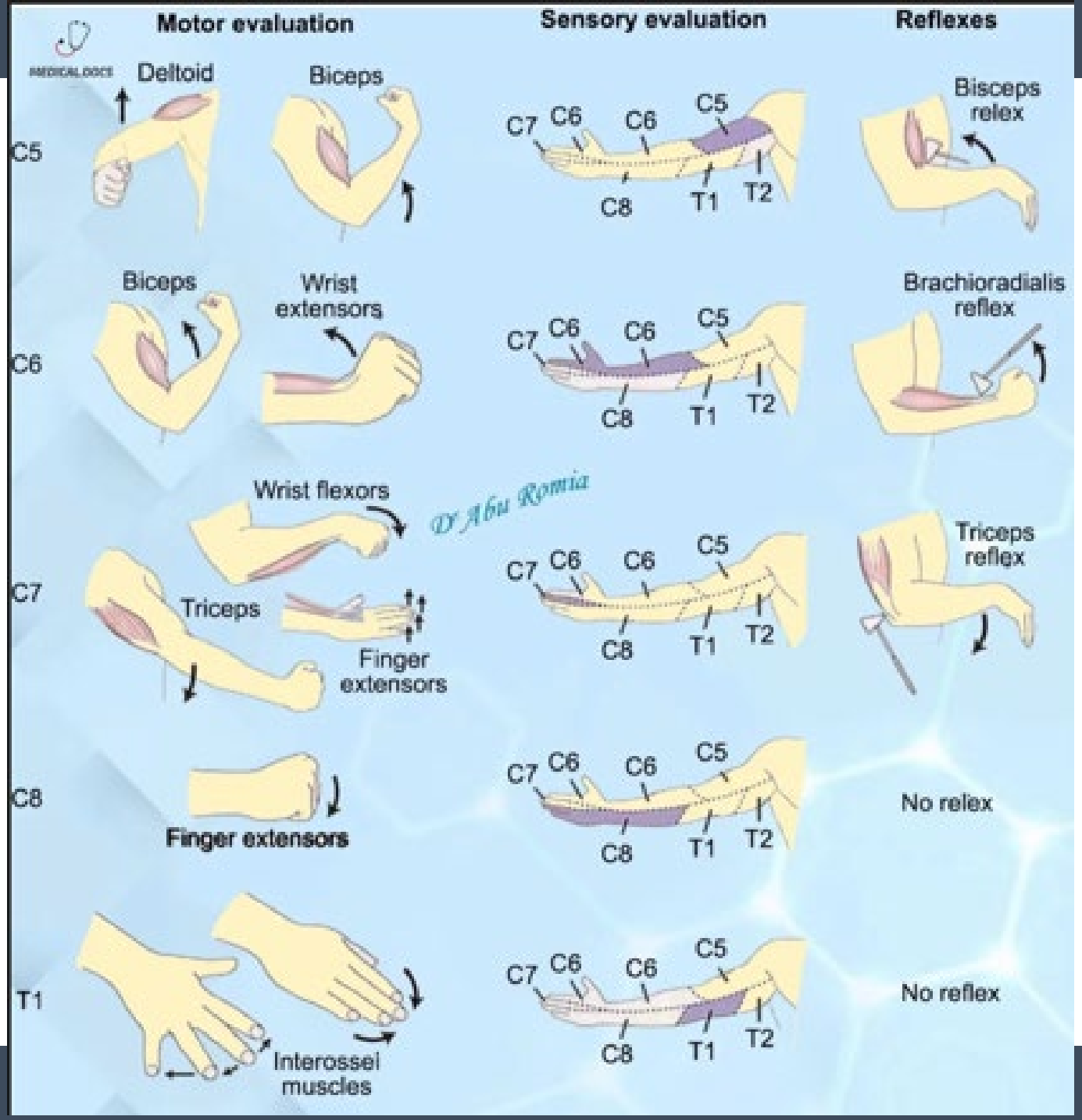


C8 Root Test



T1 Root Test

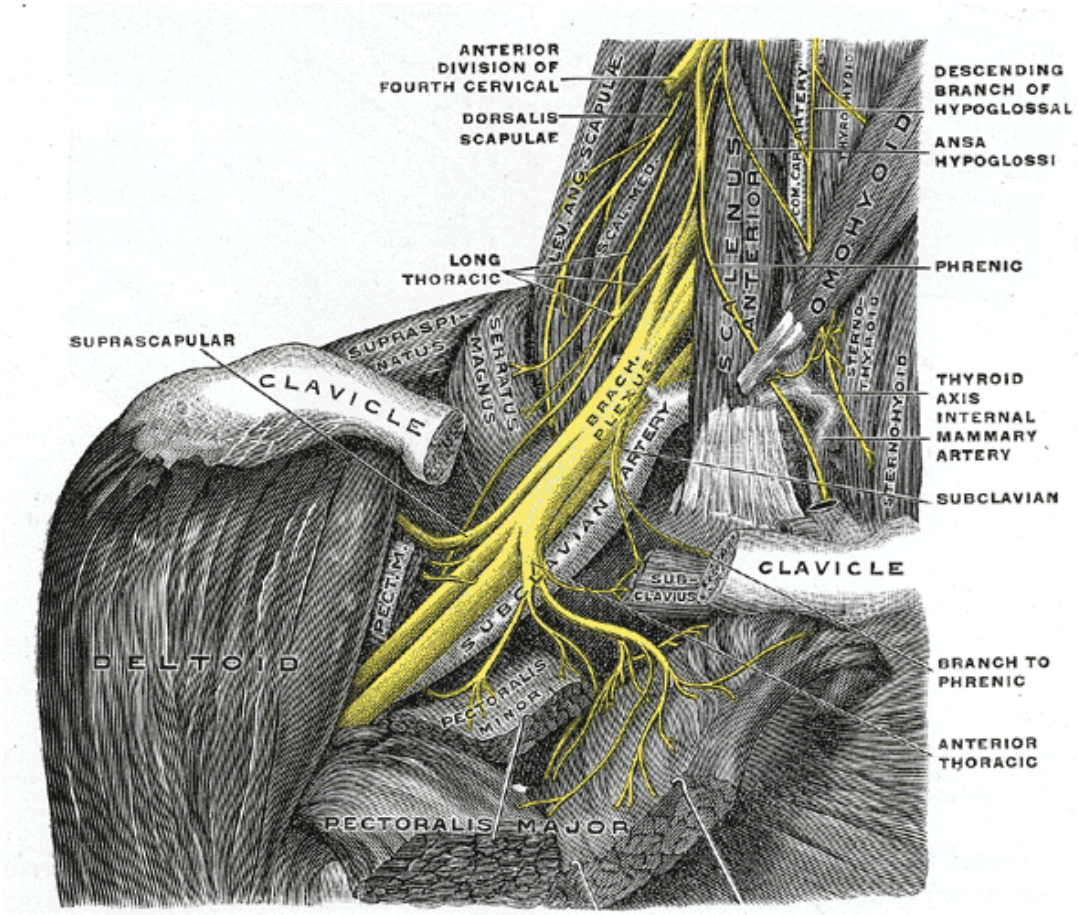


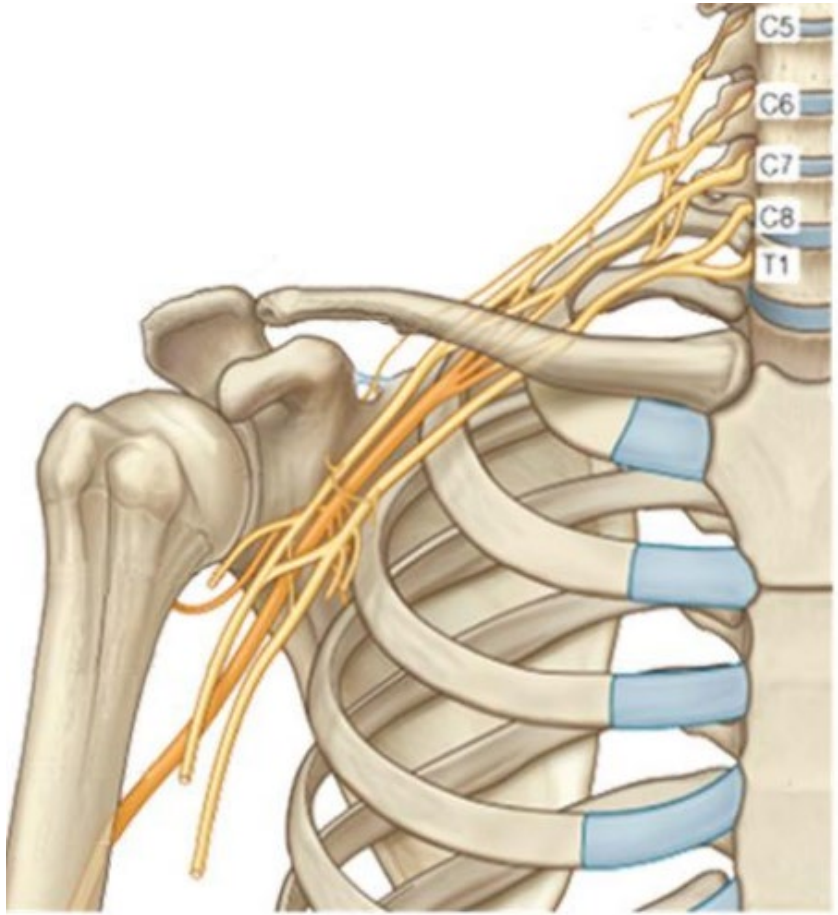

Motor evaluation
Sensory evaluation
Reflexes
Deltoid
Biceps
C5
C7 C6 C6 C5
C8 T1 T2
Bisceps relex
Biceps
Wrist extensors
C6
Brachioradialis reflex
Wrist flexors
Triceps
Finger extensors
C7
Triceps reflex
Finger extensors
C8
No relex
Interossei muscles
T1
No reflex
Dr Abu Romia

Learning Outcomes

- Give an account of the distribution of cutaneous nerves of the upper limb
- Describe the pattern of dermatome innervation of the upper limb
- Describe the myotome map of the upper limb, and list the nerves responsible for movement around each joint of the upper limb
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The Brachial Plexus

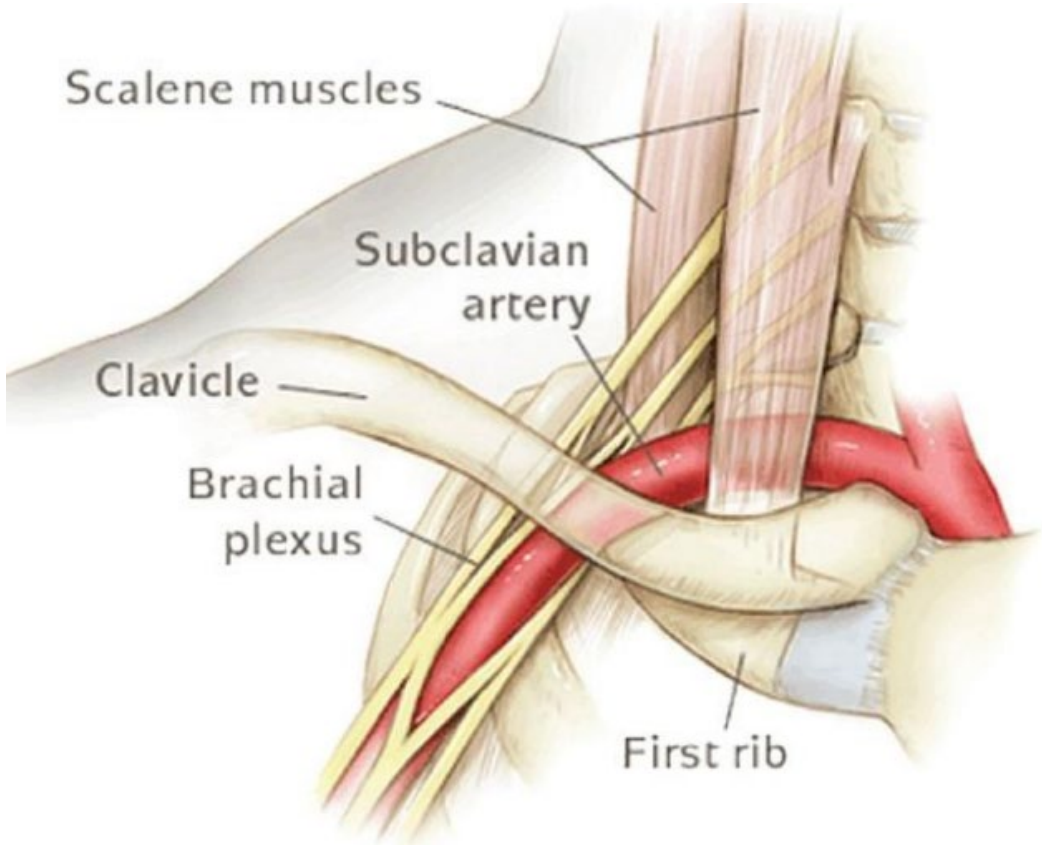




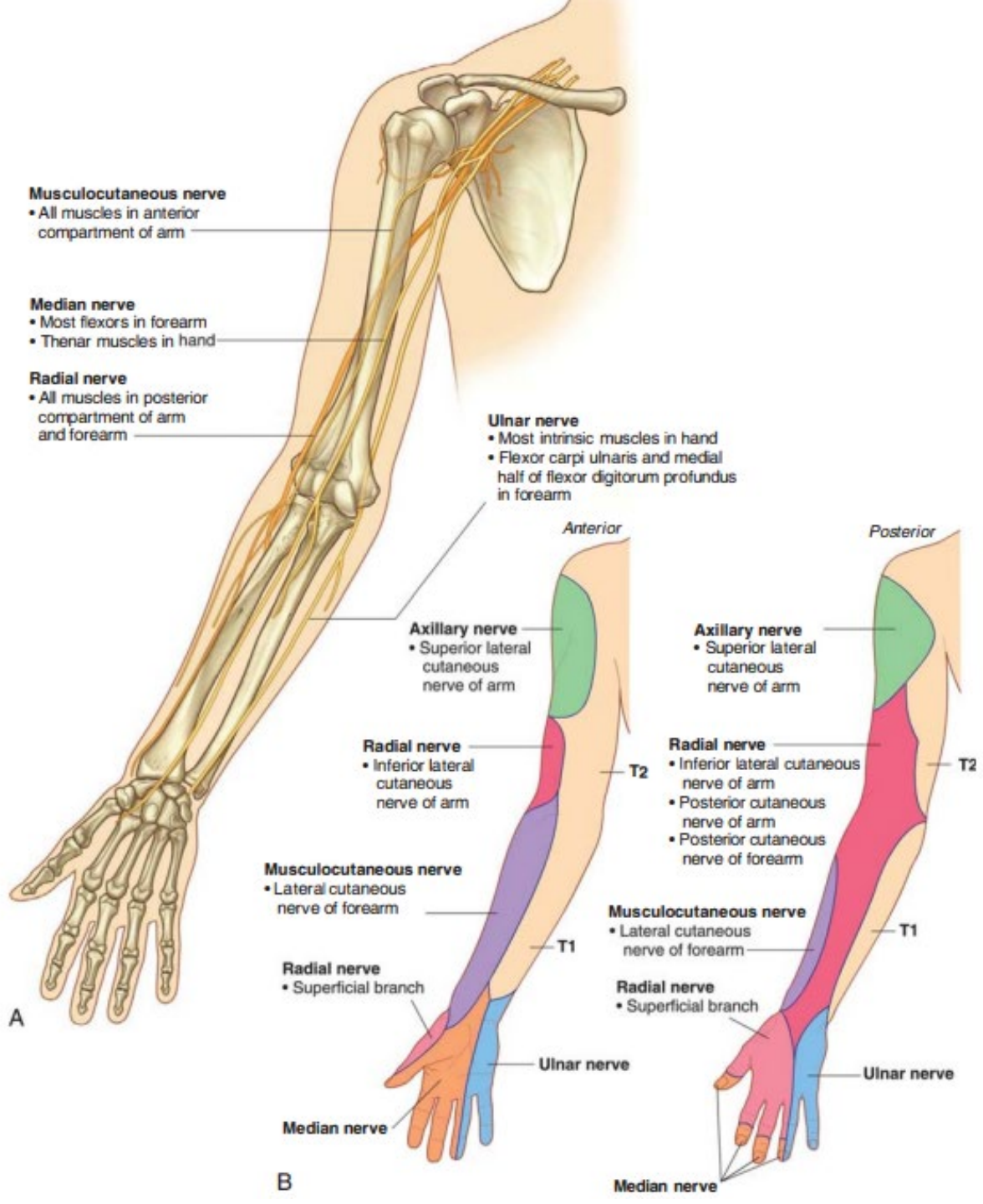
Interscalene Nerve Block



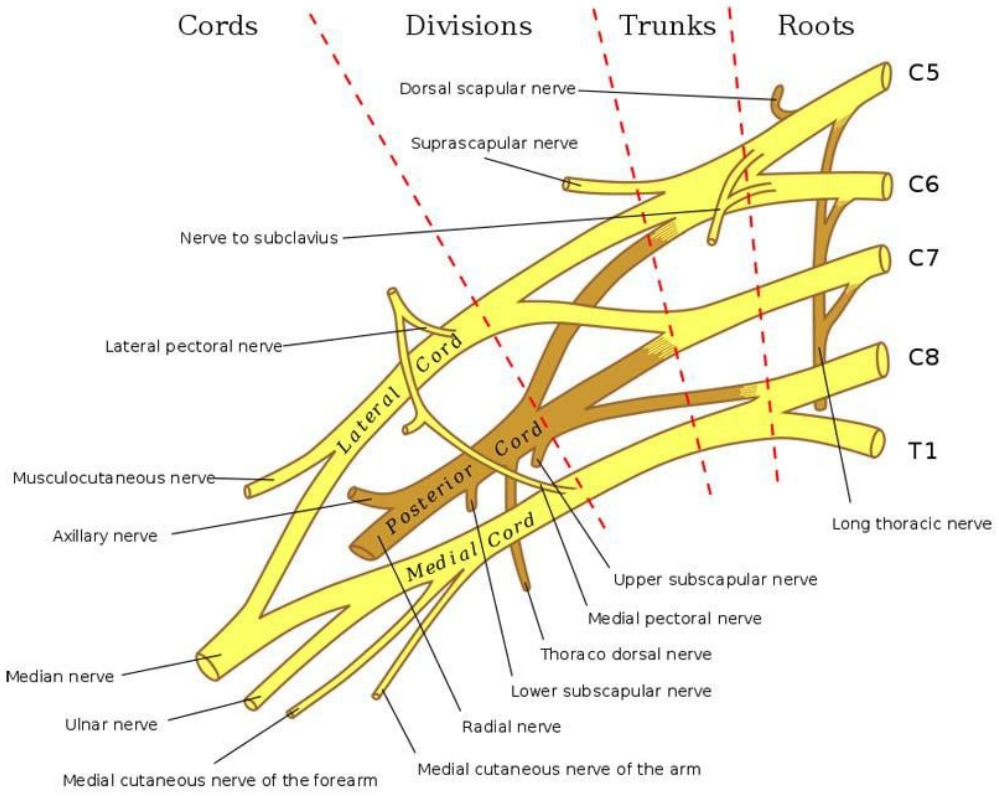




Distribution of the nerves of the Upper Limb

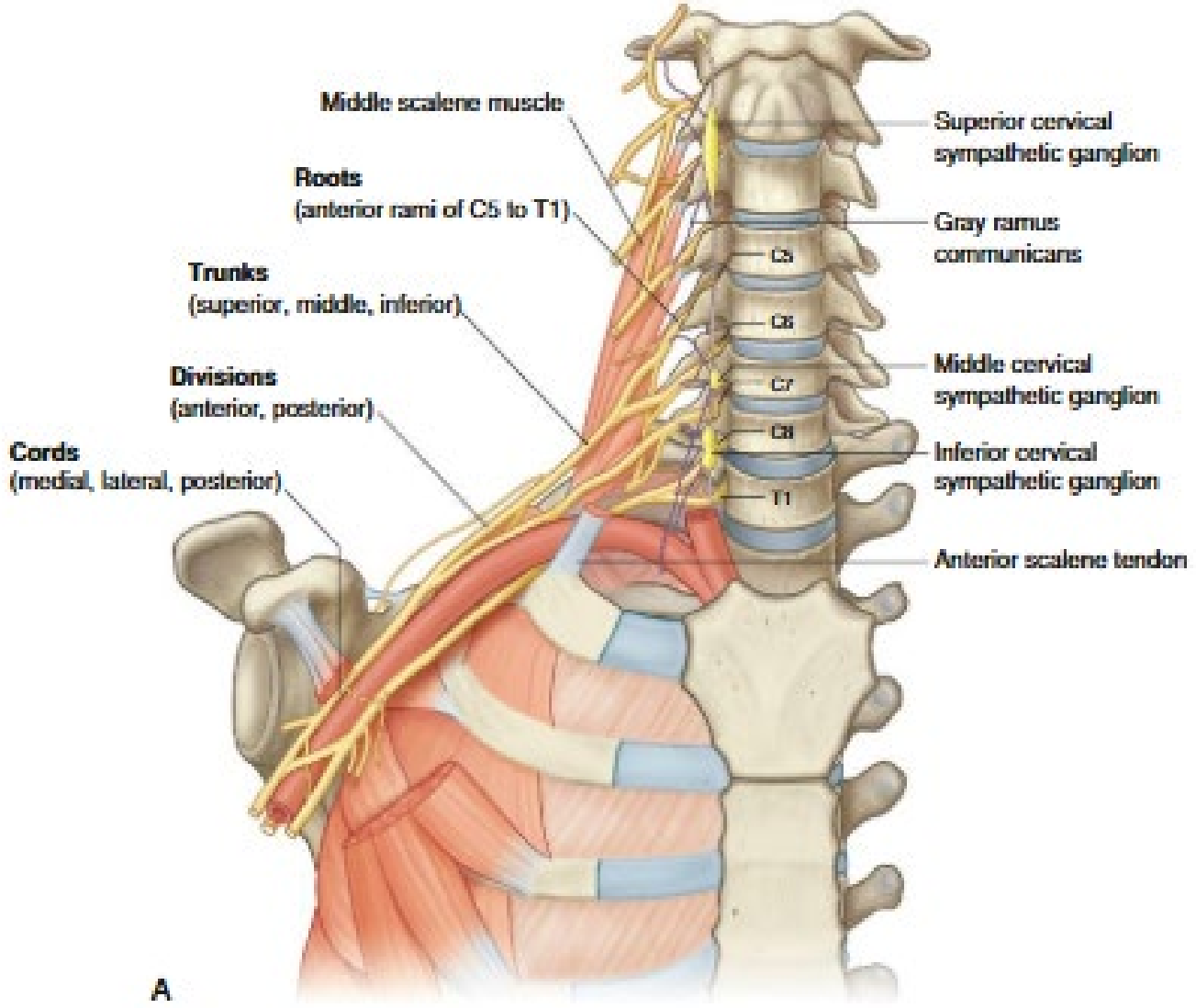


Organisation of the brachial plexus

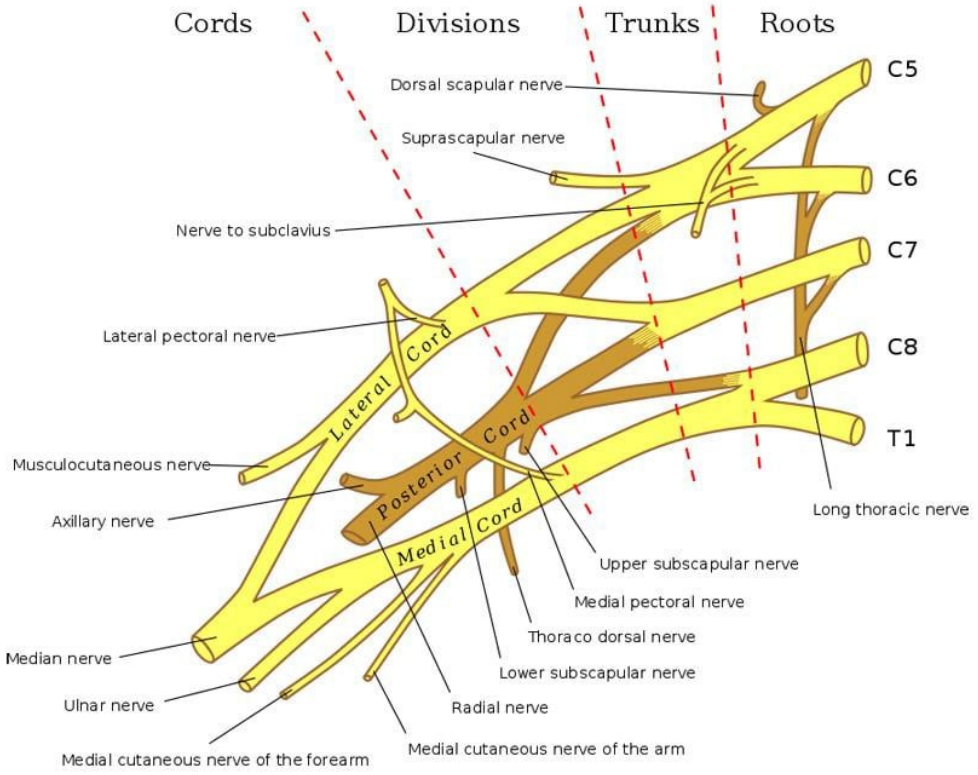


- **<u>BRACHIAL PLEXUS</u>**
  - Roots, trunks, divisions, cords and branches
- Formed by 5 **<u>ROOTS</u>**
  - Anterior rami of the lower four cervical and the first thoracic nerve roots (C5 – C8, T1)
  - They form brachial plexus after giving off their segmental supply to the muscles of the neck
  - The roots merge to form **<u>TRUNKS</u>**
- **TRUNKS**
  - Upper (C5&C6)
  - Middle (C7)
  - Lower (C8, T1)
- Each trunk divides into anterior and posterior divisions (six divisions)
- **<u>DIVISIONS</u>** re-group to form three cords (anterior, lateral and posterior (relative to their position to the axillary artery)

Brachial Plexus

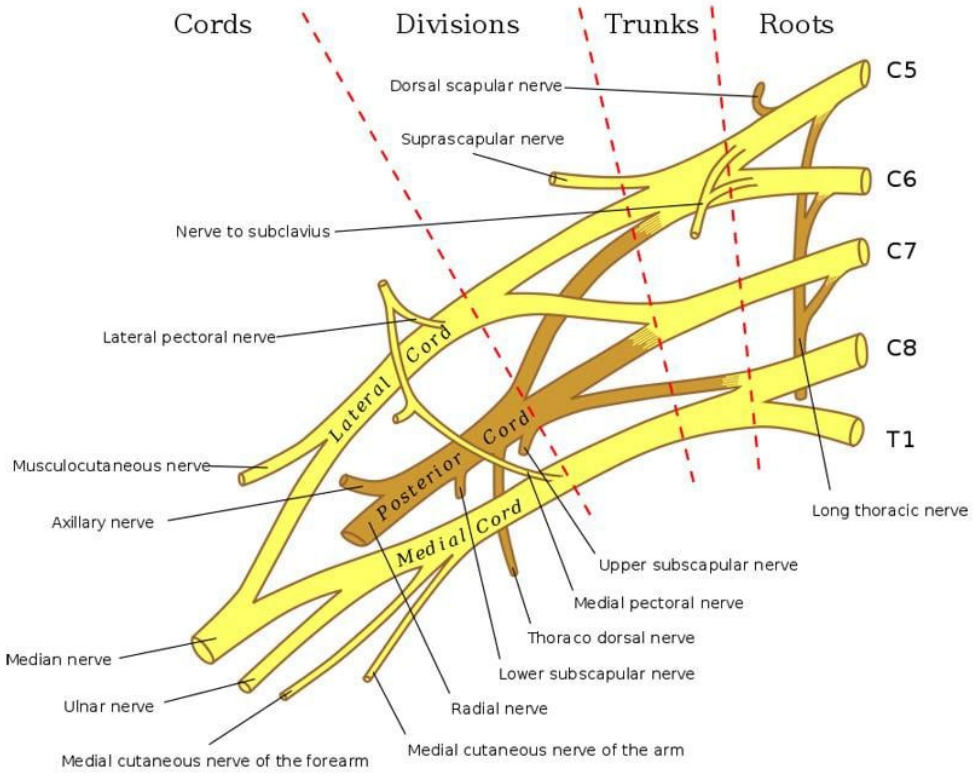


Organisation of the brachial plexus - 2

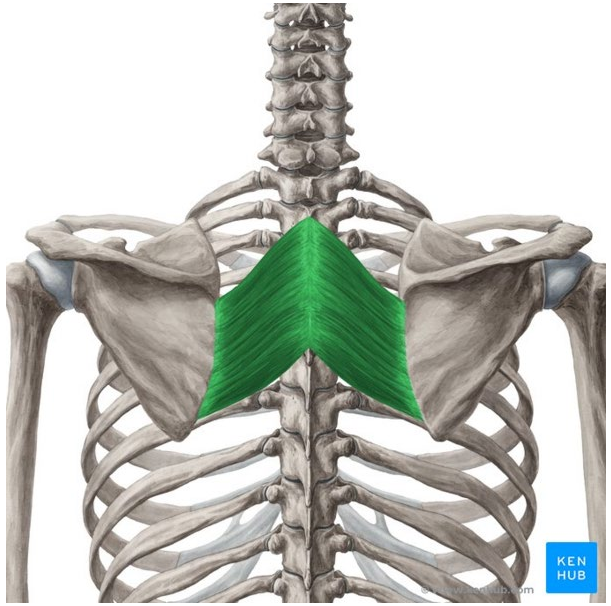


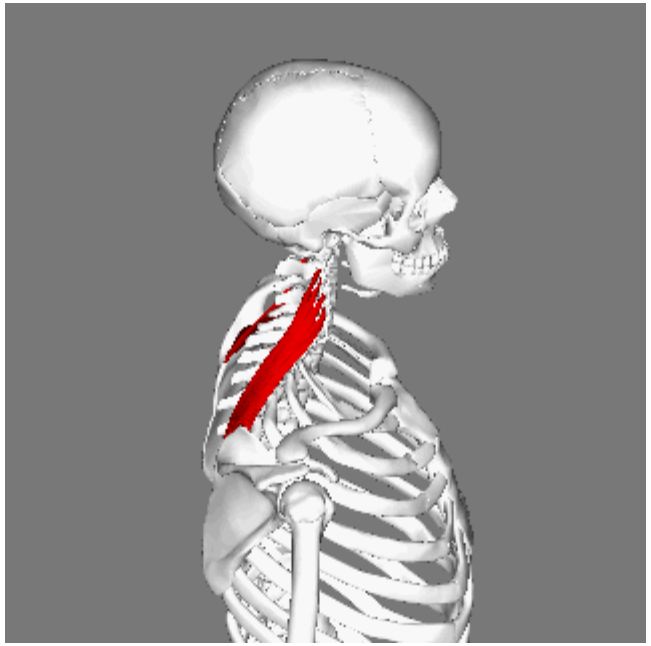
- **CORDS**
  - Posterior cord is formed from the three posterior divisions of the trunks (C5-C8, T1)
  - Lateral cord is the anterior divisions of the upper and middle trunks (C5-C7)
  - Medial Cord is the continuation of the anterior division of the lower trunk (C8, T1)

Brachial plexus branches - Roots

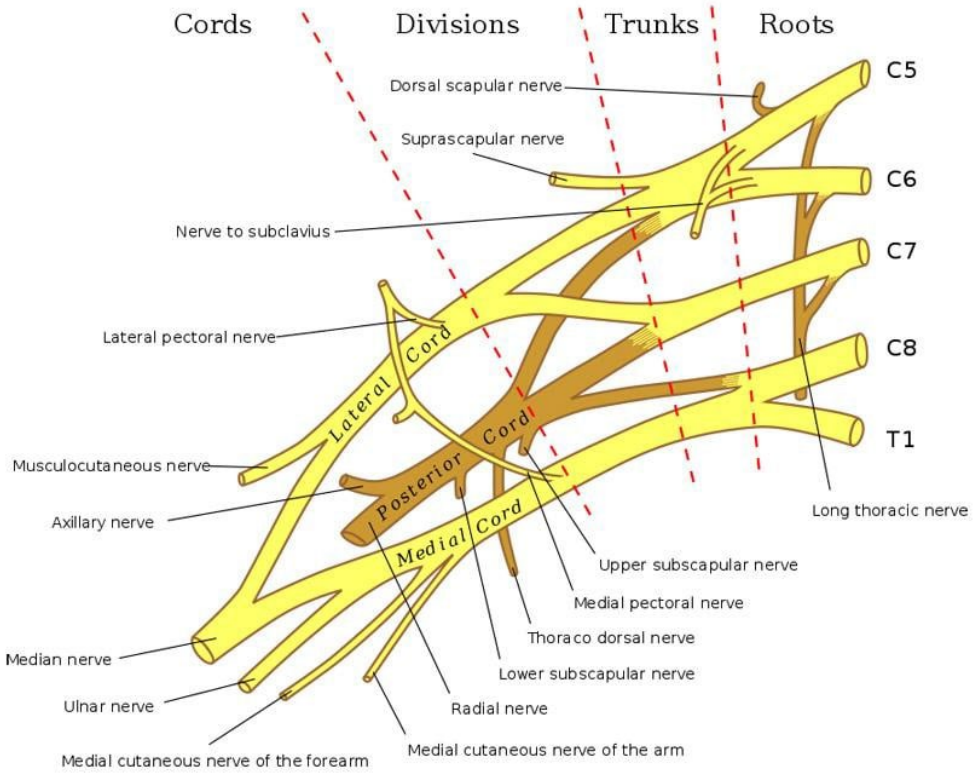


- Dorsal scapular Nerve (C4, C5)
  - Rhomboids muscles (major & minor) and levator scapulae)

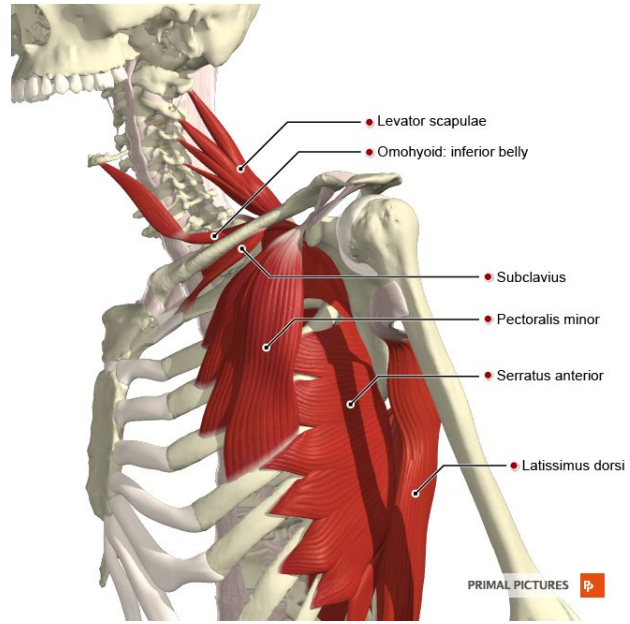




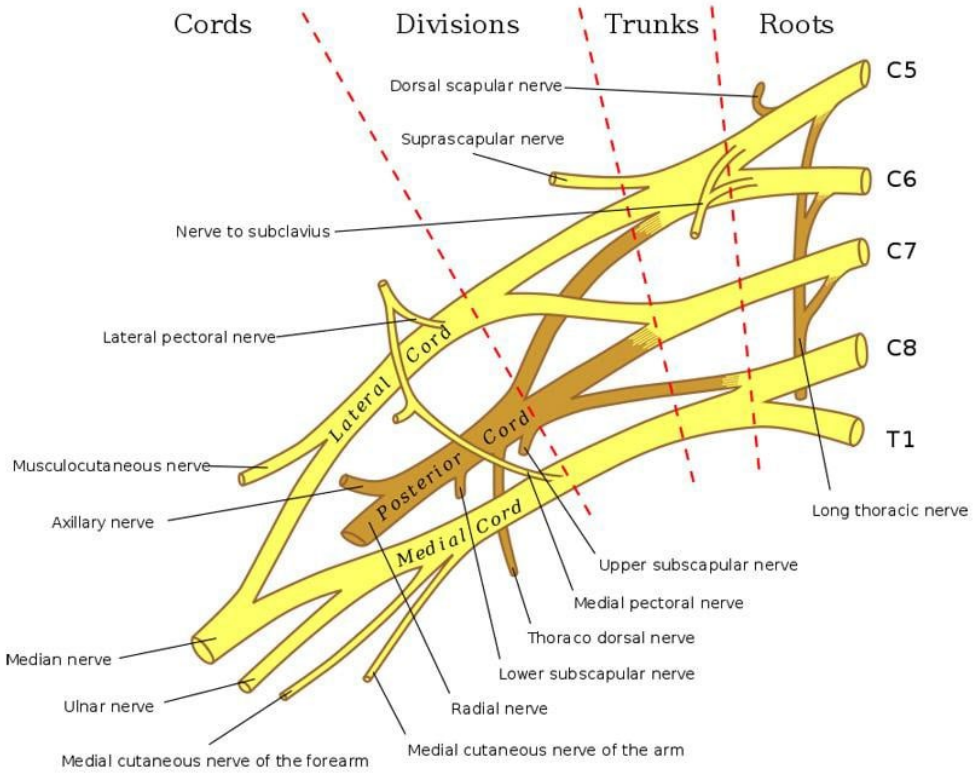
Brachial plexus branches - Roots



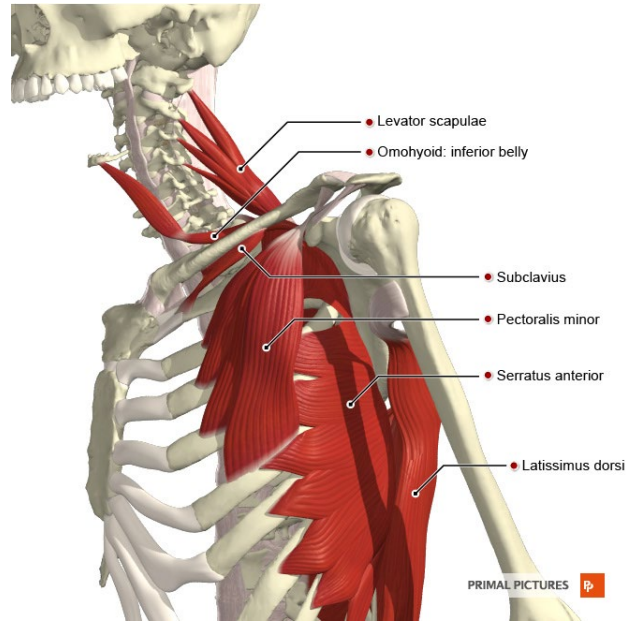
- Dorsal scapular Nerve (C4, C5)
  - Rhomboids muscle and levator scapulae)
- Long Thoracic Nerve (C5, C6, C7)
  - Serratus Anterior



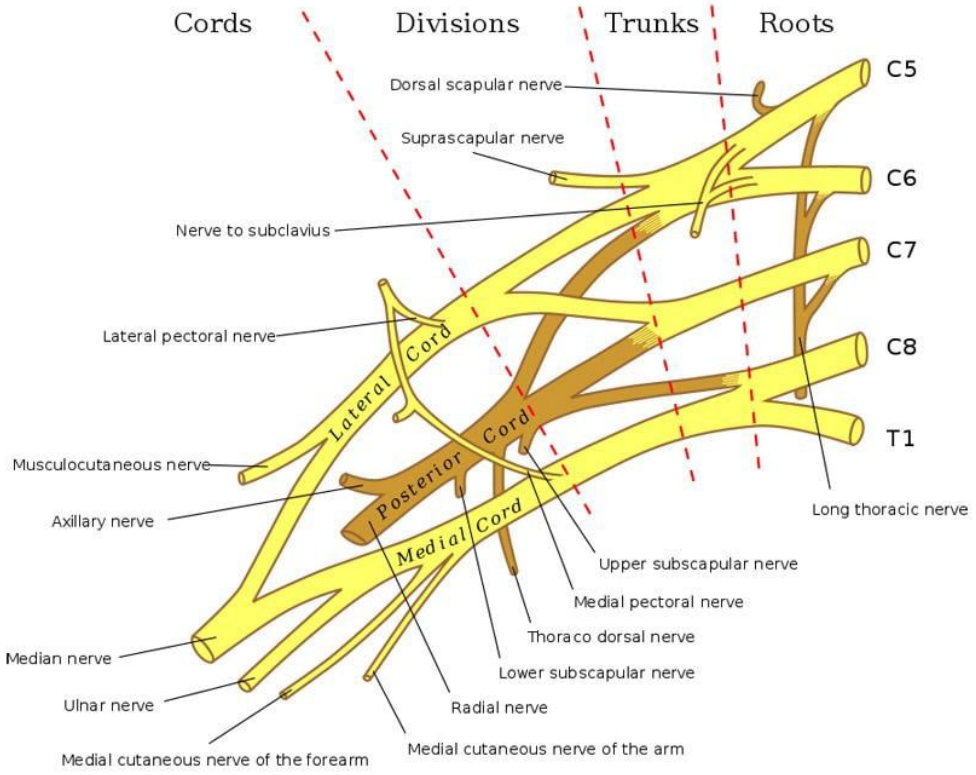
Brachial plexus branches - Roots



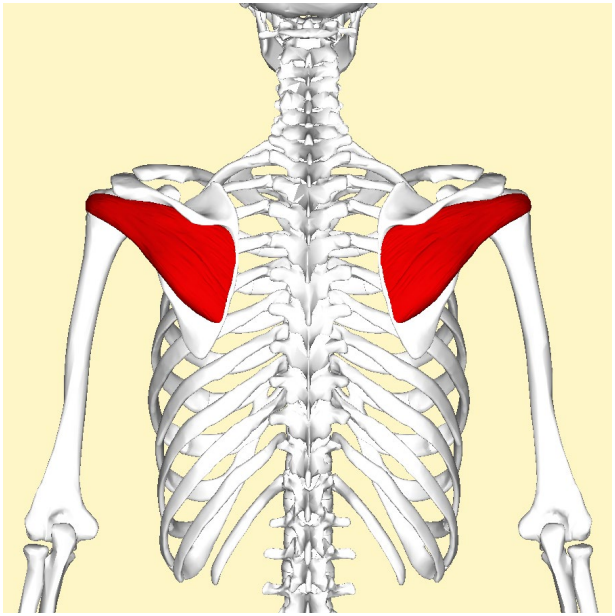
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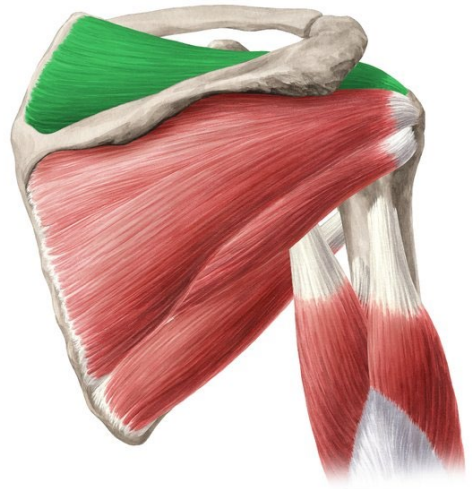


Brachial plexus branches – Upper Trunk



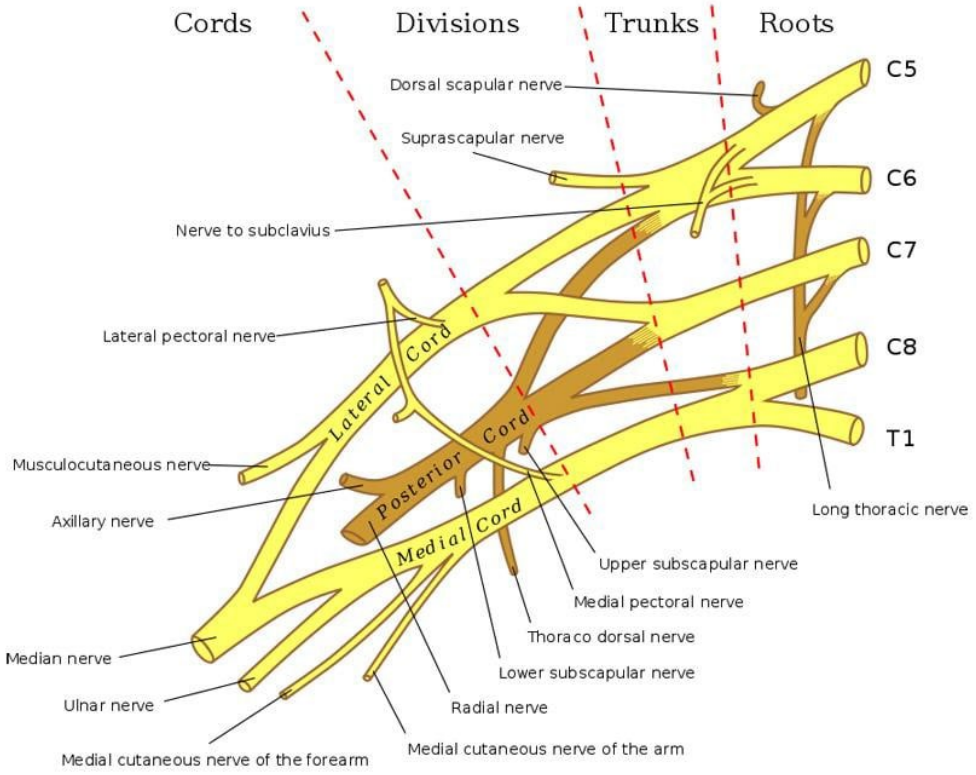
- Nerve to Subclavius muscle (C5, C6)
- Suprascapular Nerve (C5, C6)
  - Supraspinatus & Infraspinatus muscles





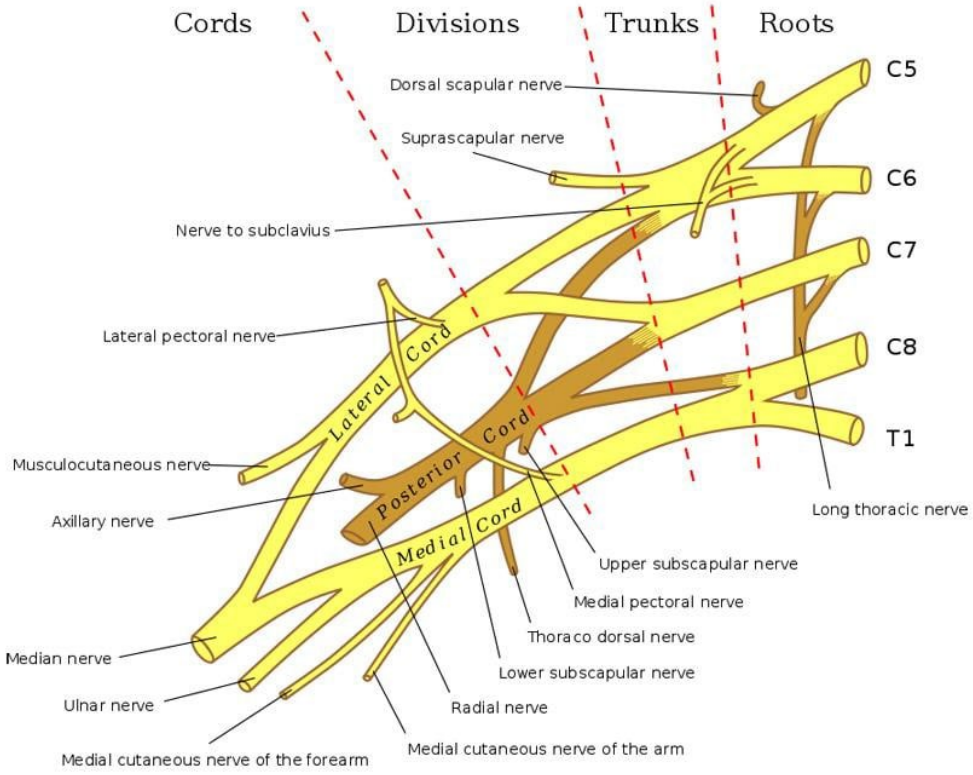
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Brachial plexus branches – Lateral cord

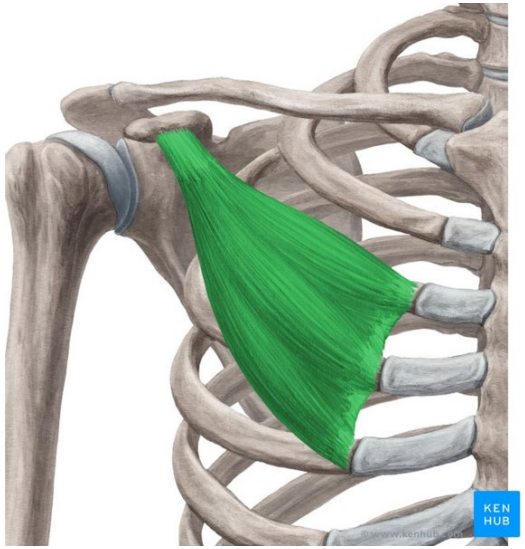


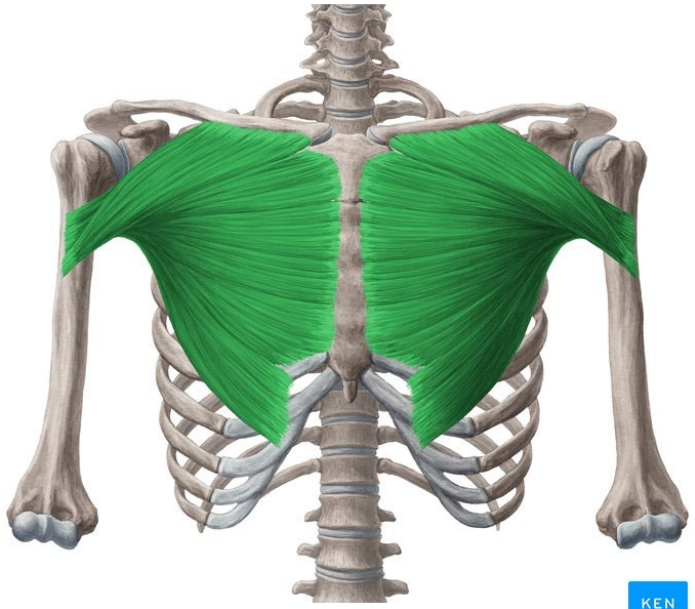
1. Lateral Pectoral Nerve (C5, C6, C7)
2. Musculocutaneous Nerve (C5, C6, C7)
3. Lateral root of the Median Nerve (C6, C7)

Brachial plexus branches – Lateral cord



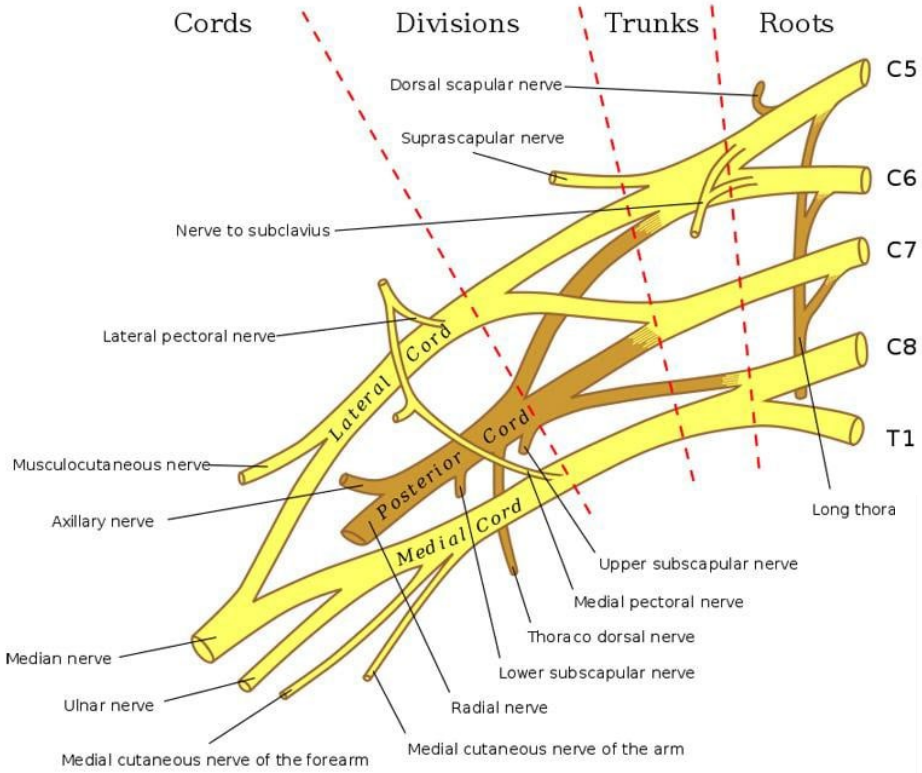
1. **Lateral Pectoral Nerve (C5, C6, C7)**
   - Pectoralis major (clavicular head) and Pectoralis minor





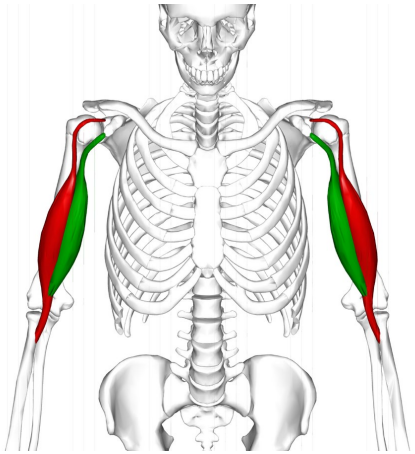
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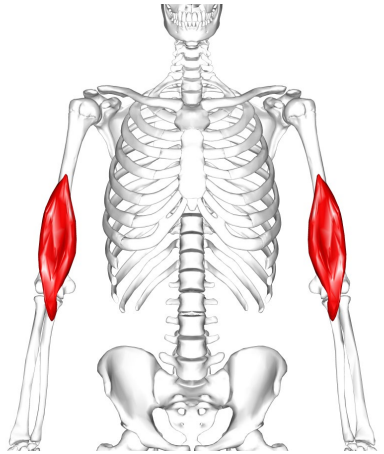
Brachial plexus branches – Lateral cord

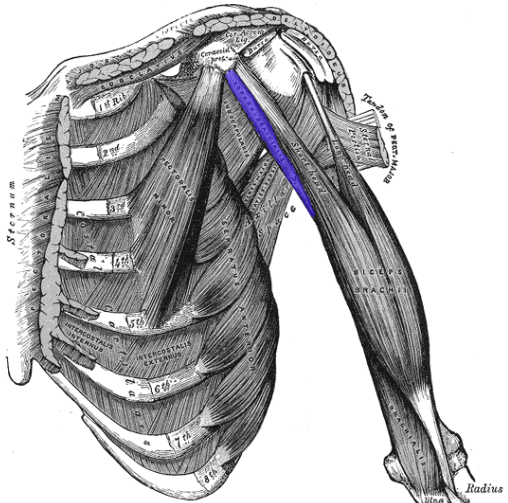


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   - Pectoralis major (clavicular head) and Pectoralis minor

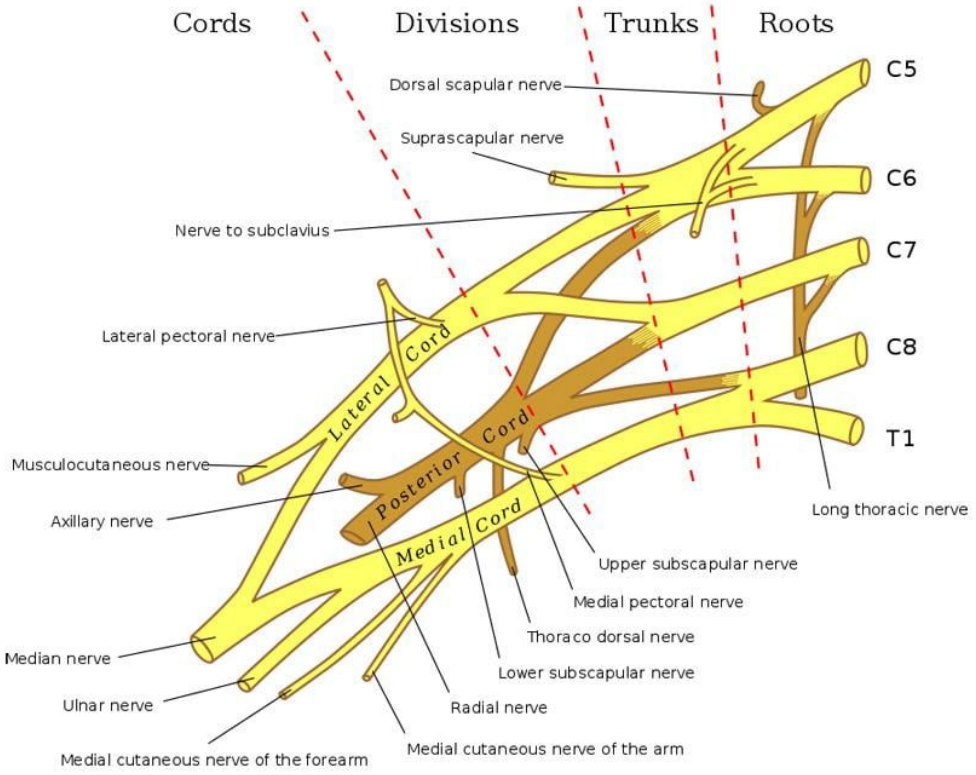
2. **Musculocutaneous nerve (C5, C6, C7)**
   - Coracobrachialis muscle, Brachialis and Biceps Brachi
   - Become the **Lateral cutaneous nerve of forearm**







Brachial plexus branches – Lateral cord

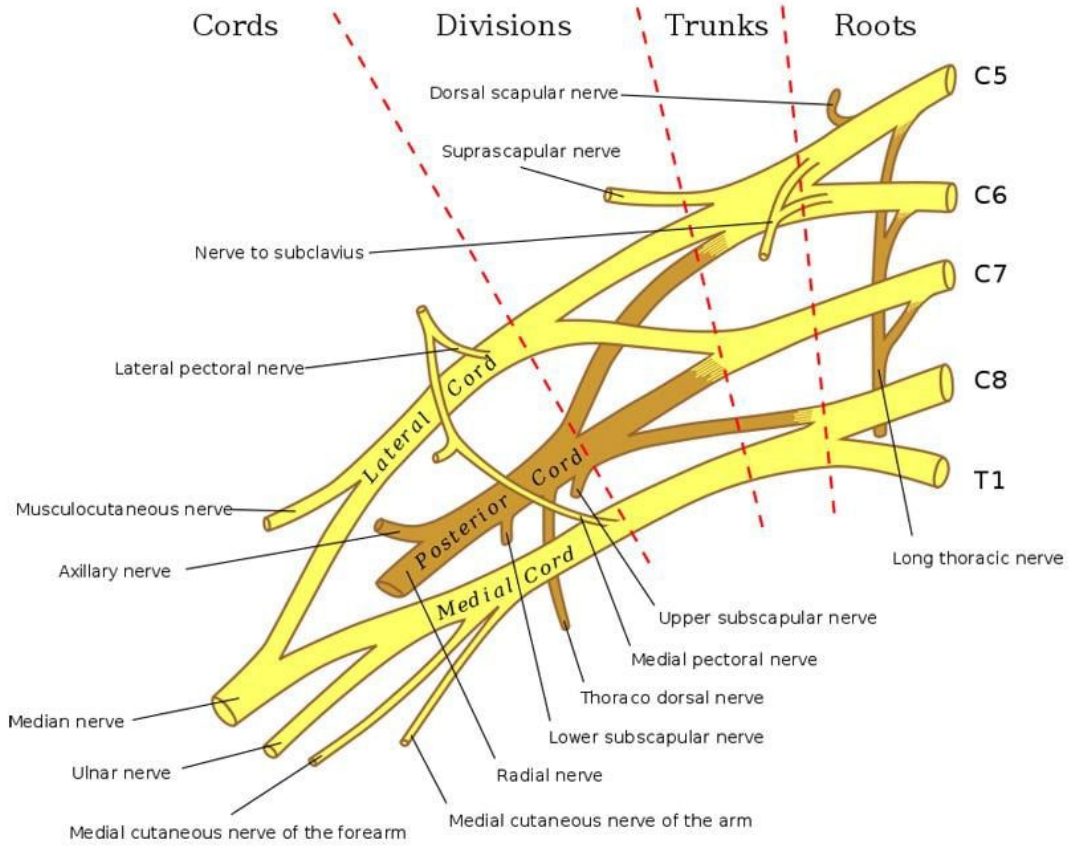


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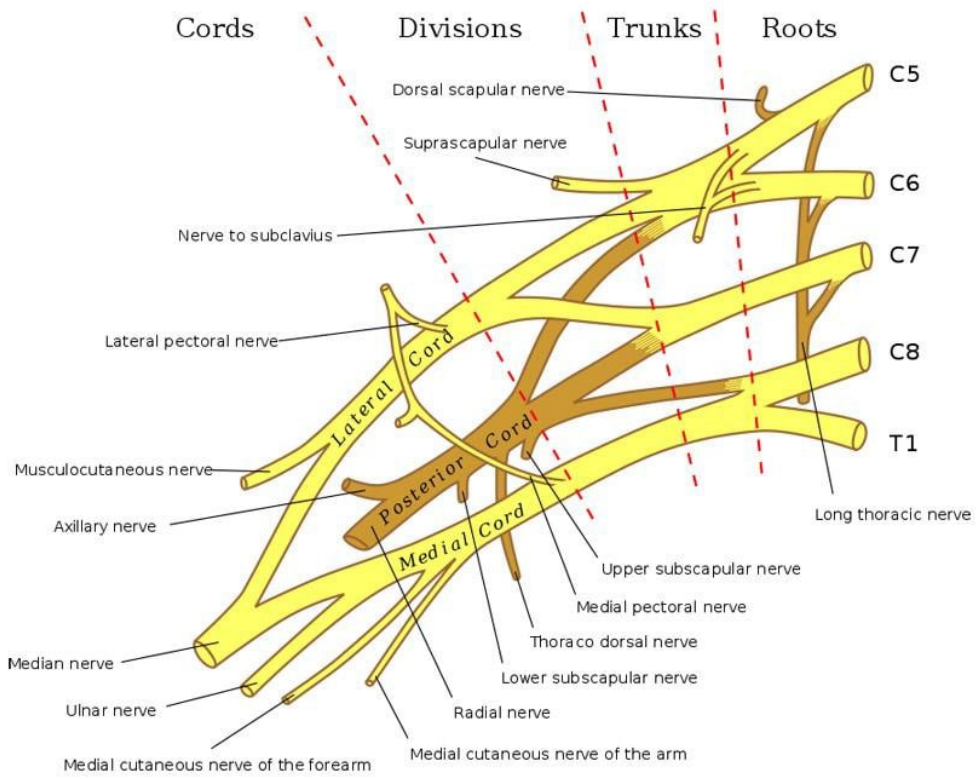
3. **<u>Lateral root of the median nerve (C6, C7)</u>**

Brachial plexus branches – Medial cord

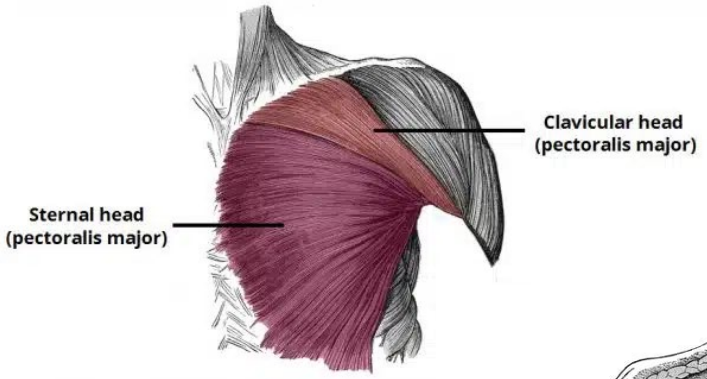


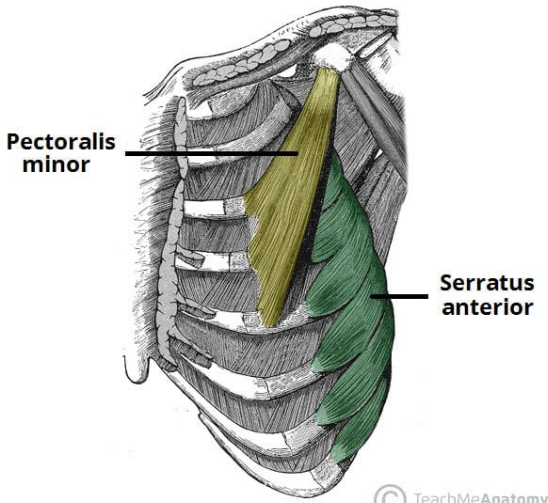
1. Medial Pectoral Nerve (C8, T1)
2. Medial cutaneous nerve of the arm (C8,T1)
3. Medial cutaneous nerve of the forearm (C8, T1)
4. Medial root of the median nerve (C8 & T1)
5. Ulnar Nerve (C8 & T1)

Brachial plexus branches – Medial cord



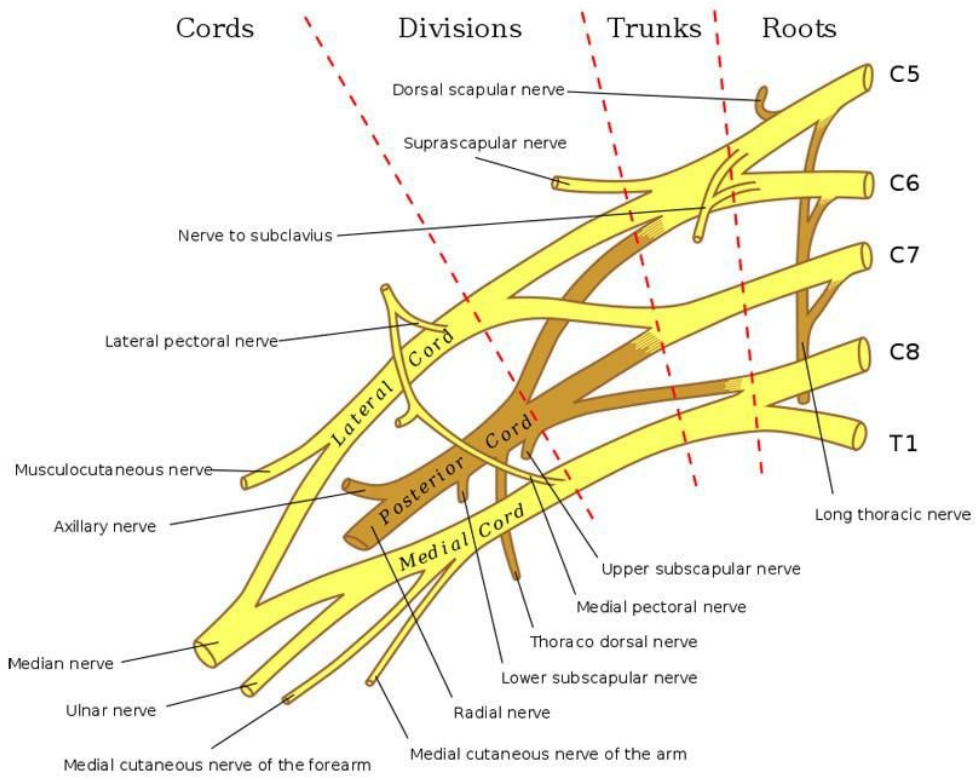
1. **Medial Pectoral Nerve (C8, T1)**
   - Pectoralis major (sternocostal and abdominal part) and Pectoralis Minor



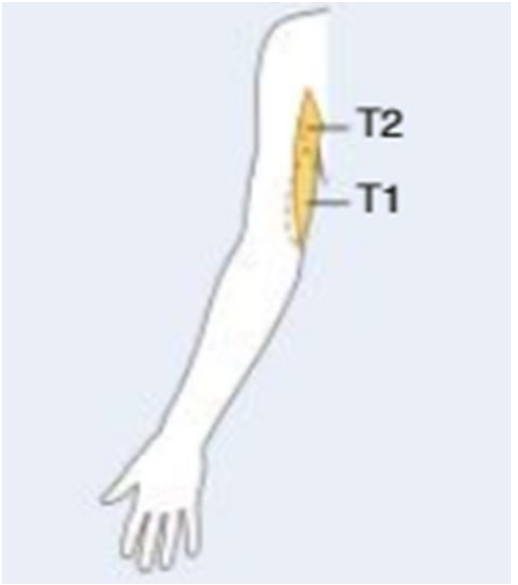


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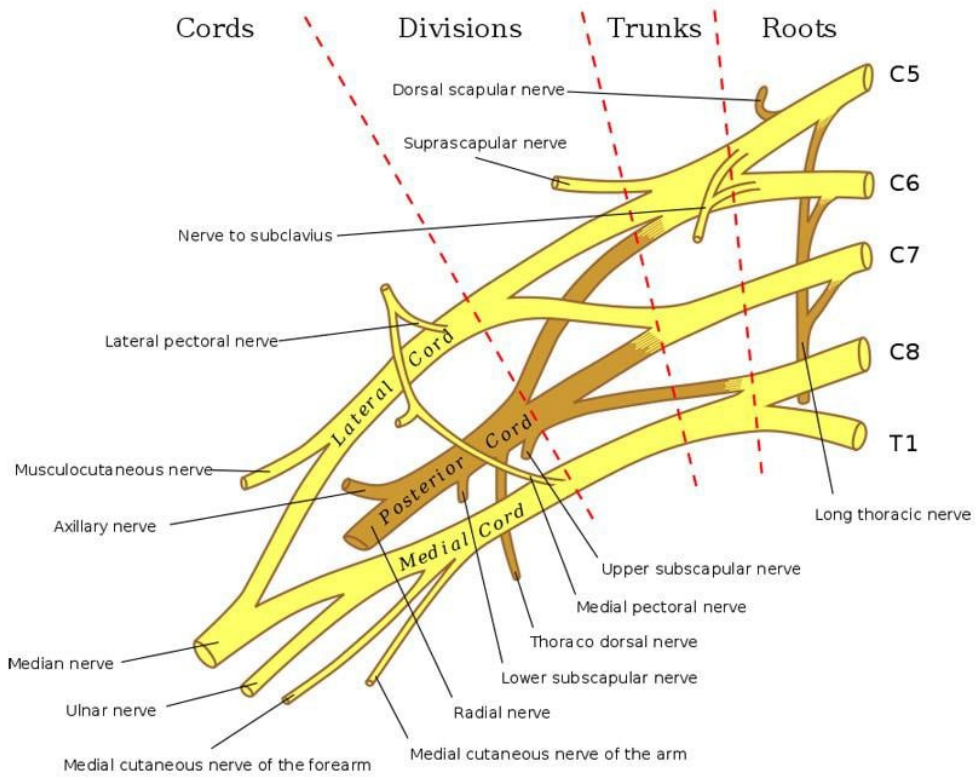
Brachial plexus branches – Medial cord



1. **Medial Pectoral Nerve (C8, T1)**
    1. Pectoralis major (sternocostal and abdominal part) and Pectoralis Minor
2. **Medial cutaneous nerve of the arm (C8,T1)**



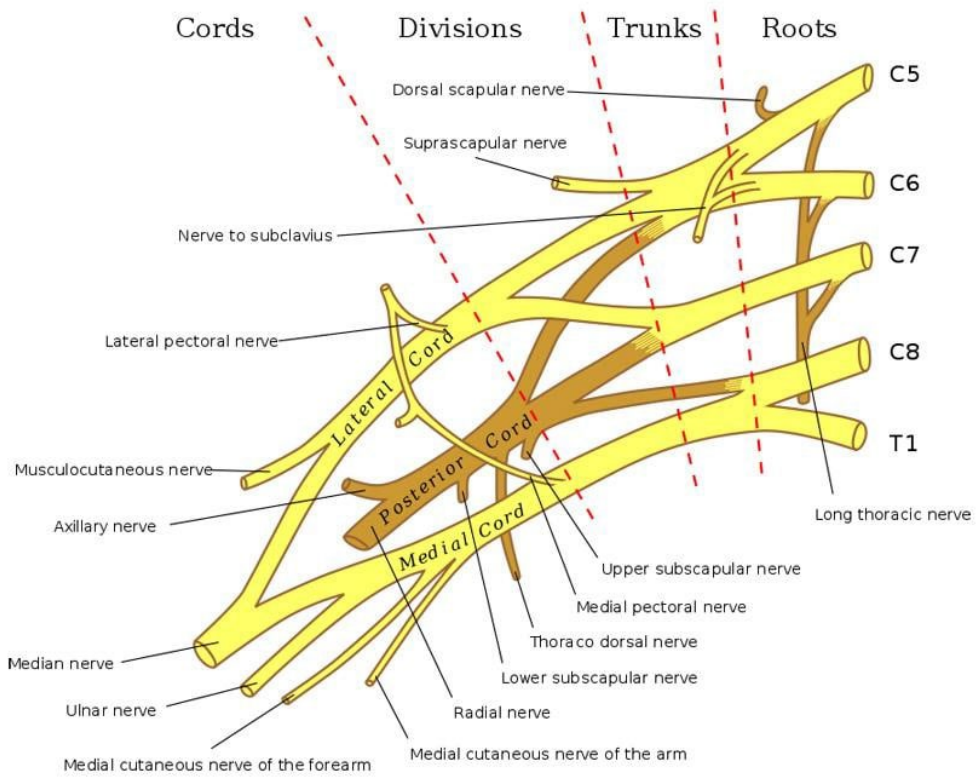
Brachial plexus branches – Medial cord



1. Medial Pectoral Nerve (C8, T1)
2. Medial cutaneous nerve of the arm (C8,T1)
3. **Medial cutaneous nerve of the forearm (C8, T1)**

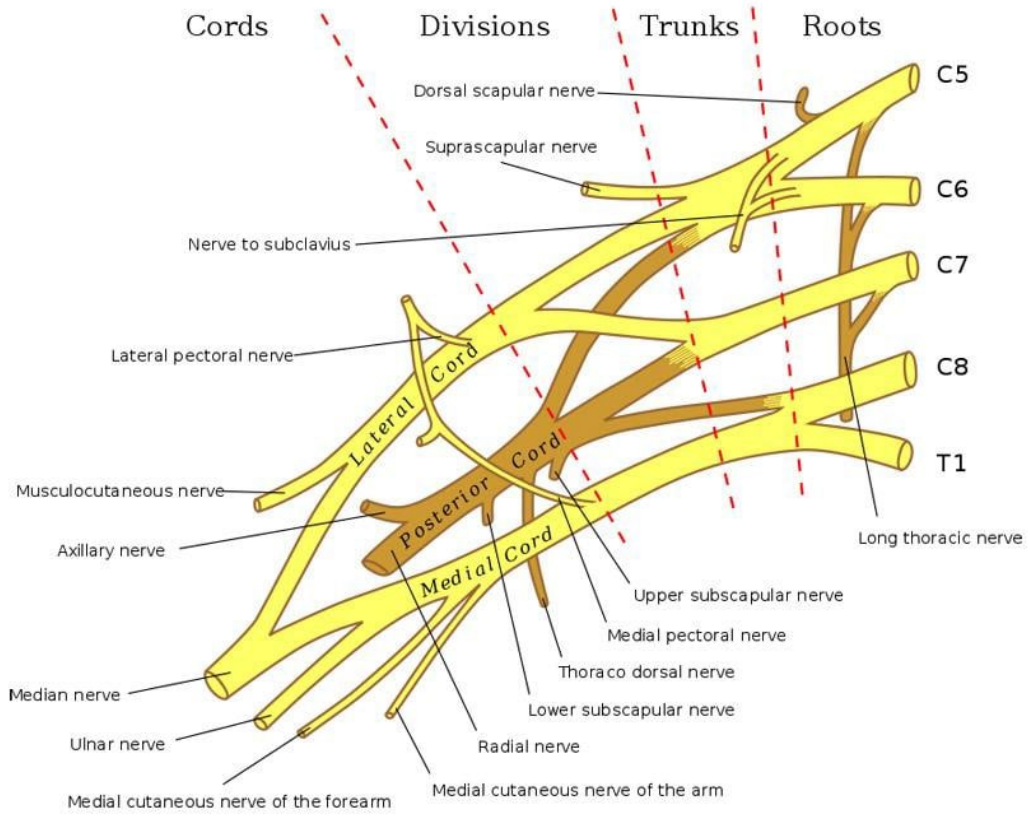


Brachial plexus branches – Medial cord



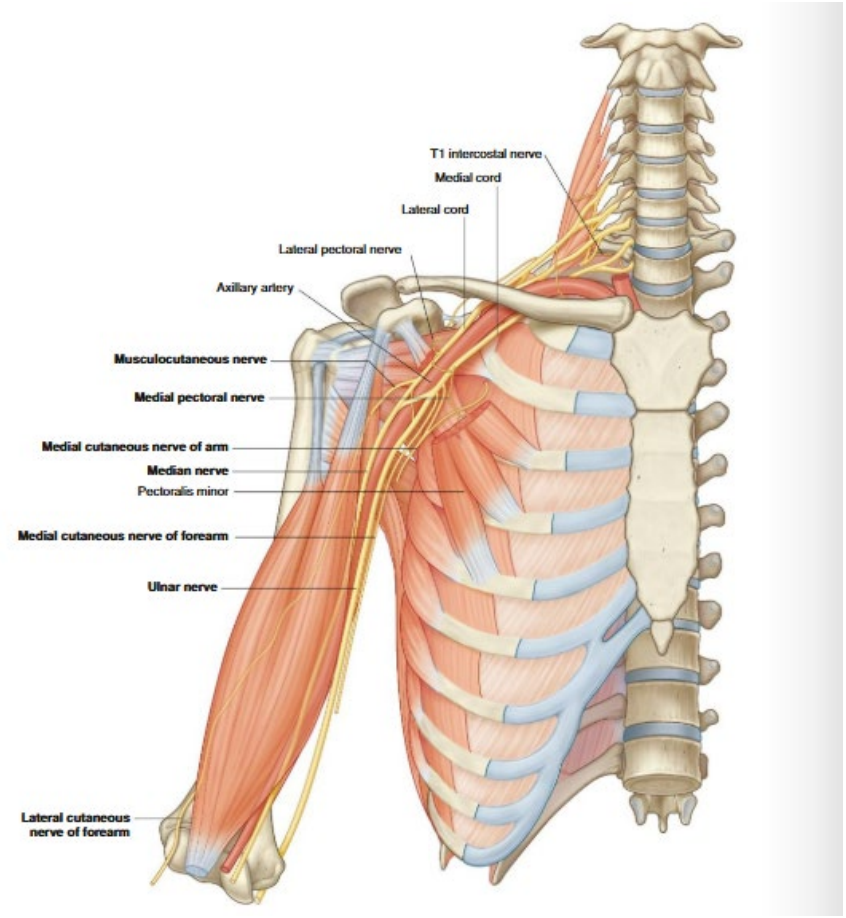
1. Medial Pectoral Nerve (C8, T1)
2. Medial cutaneous nerve of the arm (C8,T1)
3. Medial cutaneous nerve of the forearm (C8, T1)
4. **Medial root of the median nerve (C8 & T1)**
   1. joins the lateral part to form the median nerve

Brachial plexus branches – Medial cord



1. Medial Pectoral Nerve (C8, T1)
2. Medial cutaneous nerve of the arm (C8,T1)
3. Medial cutaneous nerve of the forearm (C8, T1)
4. Medial root of the median nerve (C8 & T1)
5. **Ulnar Nerve (C7, C8 & T1)**
   - Flexor carpi ulnaris,
   - Medial belies of the flexor digitorum profundus,
   - All intrinsic hand muscles except the thenar muscles and the two most lateral lumbricals
   - Cutaneous sensations
     - Little finger, half of the ring finger
     - Associated palm and wrist
     - Dorsal surface of medial part of hand

Median Nerve (C5,C6,C7,C8 & T1)



- Formed on the anterior part of the last part of axillary artery
- Passes through the arm to access the forearm
- Supplies mostly:
  - Muscles of the forearm compartment except
    - Flexor carpi ulnaris
    - Medial half of flexor digitorum profundus
  - Hand, it supplies:
    - Thenar muscles:
    - Two lateral lumbricals (index and middle finger
    - <u>Cutaneous:</u>
      - the palmar lateral three and half fingers
      - Lateral surface of the palm and wrist



Coffee break

PT4054-SEM2|PT7001-YEAR

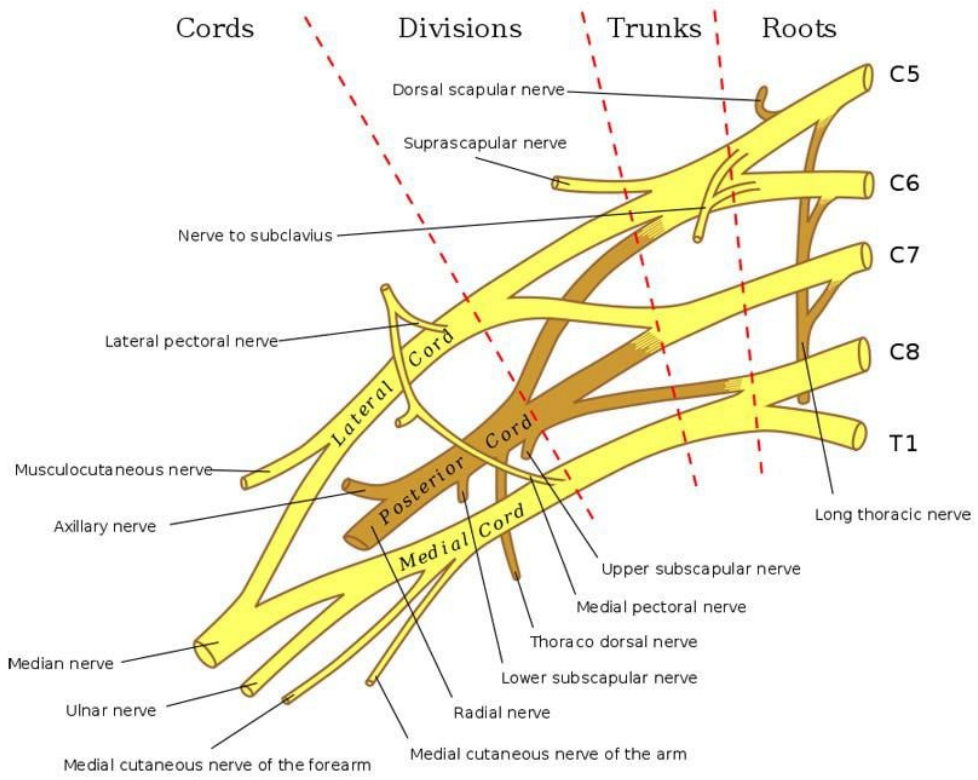
Applied Science 2, Applied Sciences

March 17, 2025 9:00 - 13:00

Lecture Physical

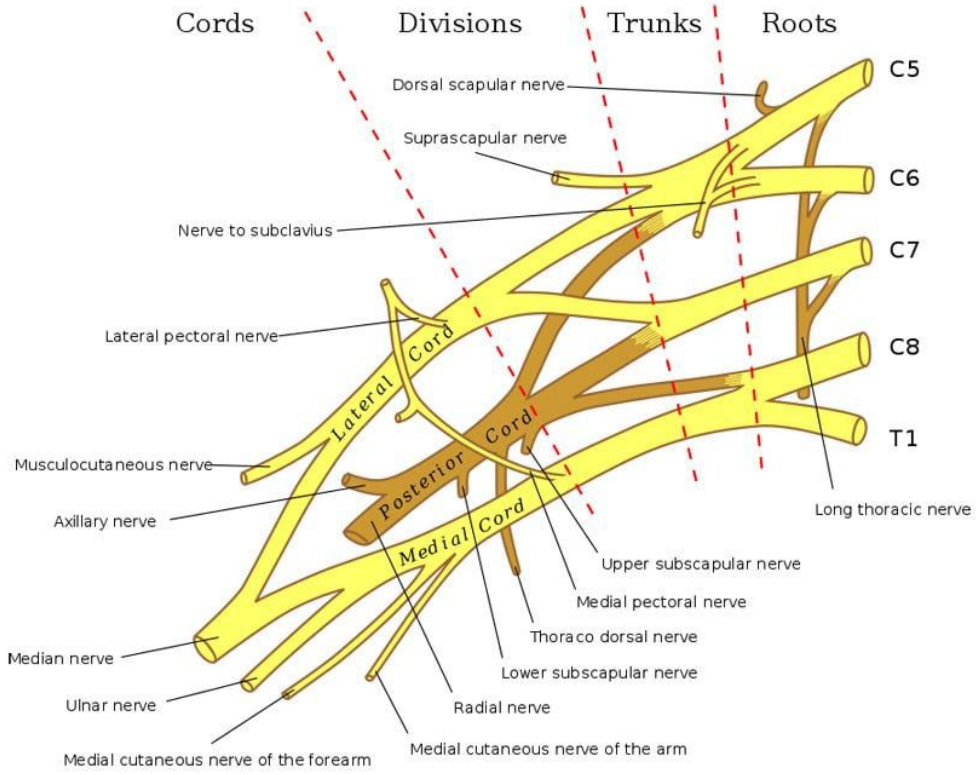
534 187

Brachial plexus branches – Posterior cord

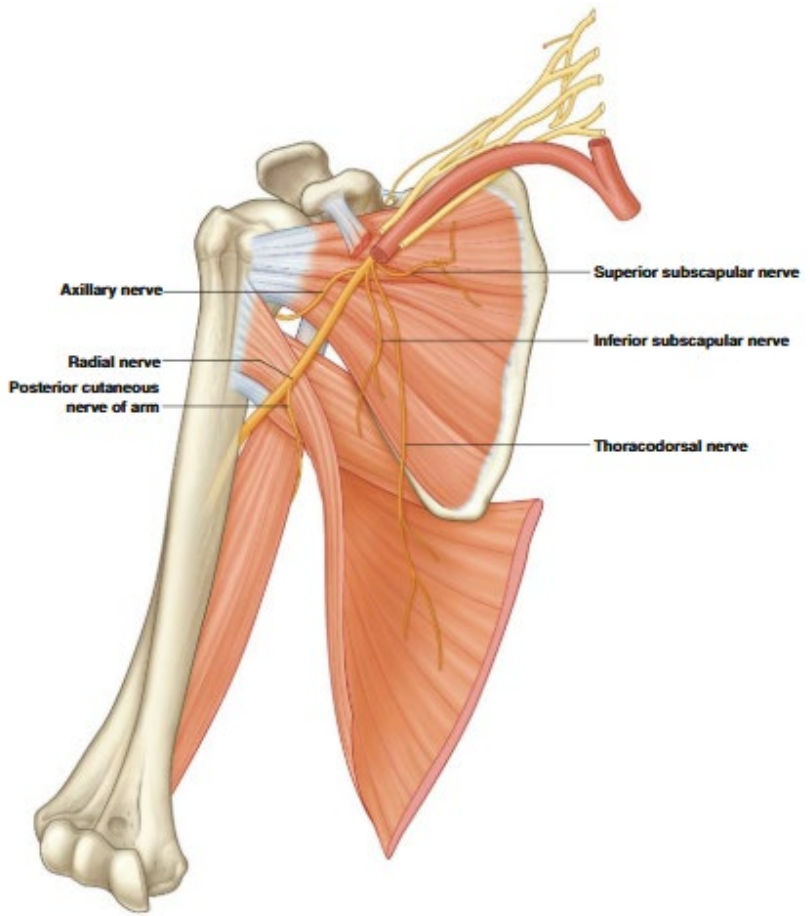


1. Upper subscapular Nerve (C5, C6)
2. Thoracodorsal nerve (C6, C7, C8)
3. Lower subscapular nerve (C5,C6)
4. Axillary nerve (C5,C6)
5. Radial Nerve (C5,C6,C7,C8 & T1)

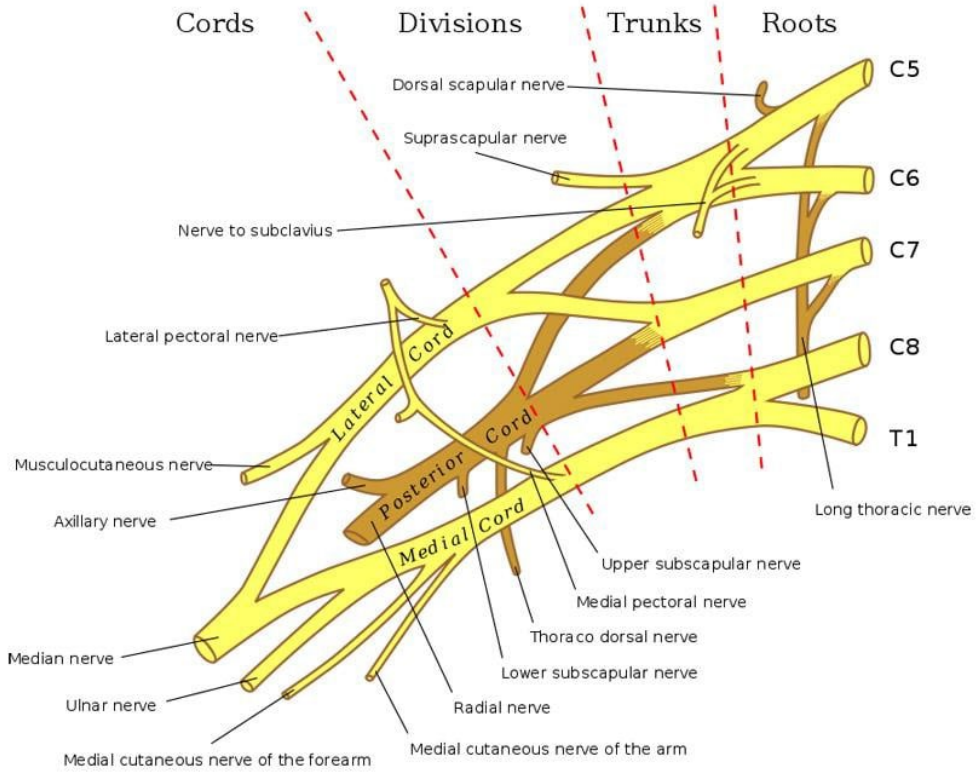
Brachial plexus branches – Posterior cord



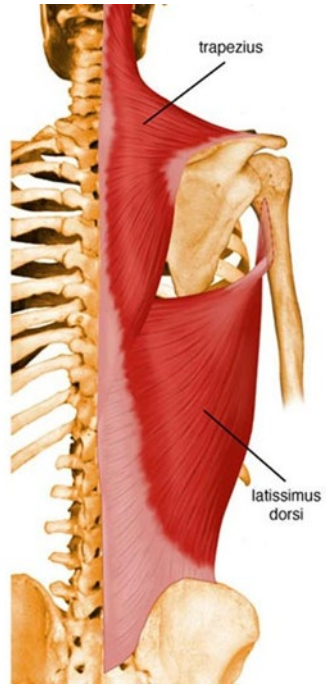
1. Upper subscapular Nerve (C5, C6)
   - Subscapularis (upper fibres)



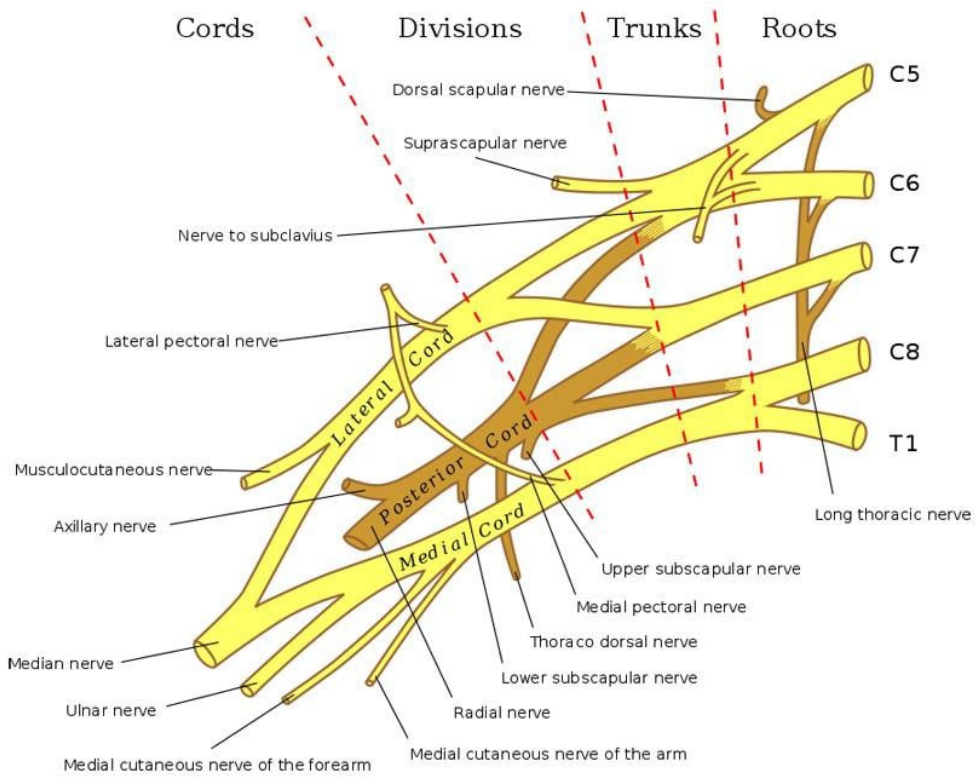
Brachial plexus branches – Posterior cord



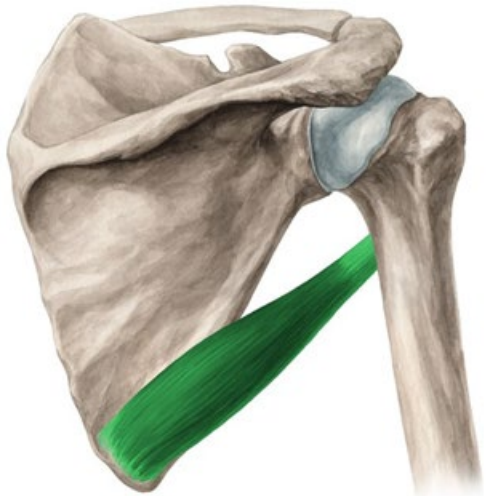
1. Upper subscapular Nerve (C5, C6)
2. Thoracodorsal nerve (C6, C7, C8)
   - Latissimus dorsi



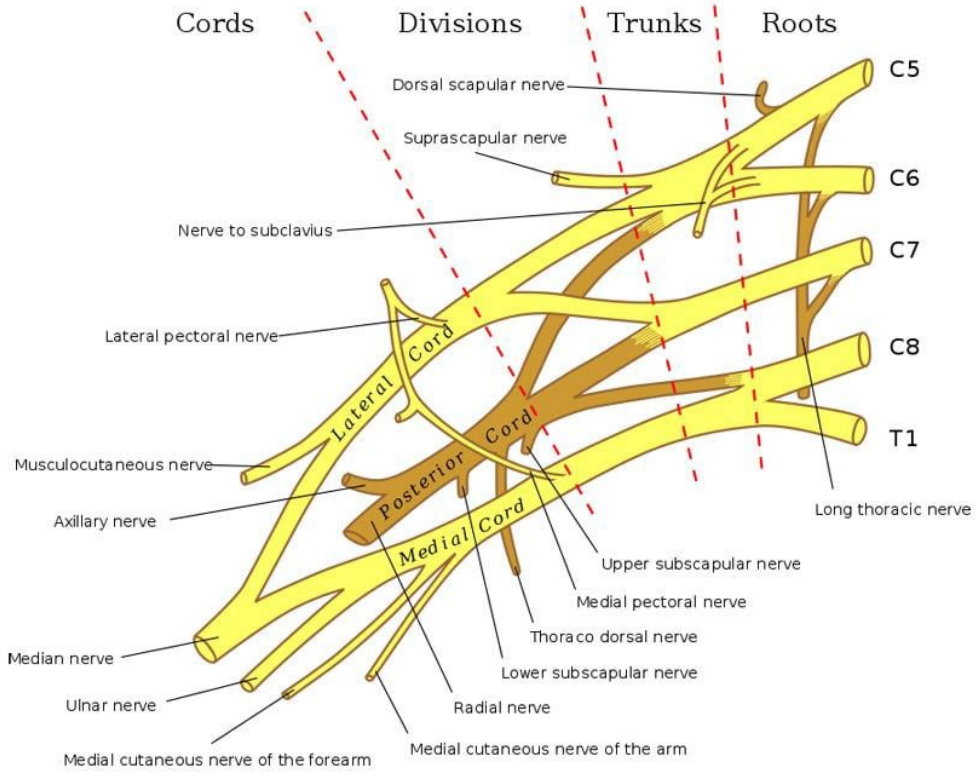
Brachial plexus branches – Posterior cord



1. Upper subscapular Nerve (C5, C6)
2. Thoracodorsal nerve (C6, C7, C8)
3. Lower subscapular nerve (C5,C6)
   - Subscapularis (lower parts)
   - Teres major

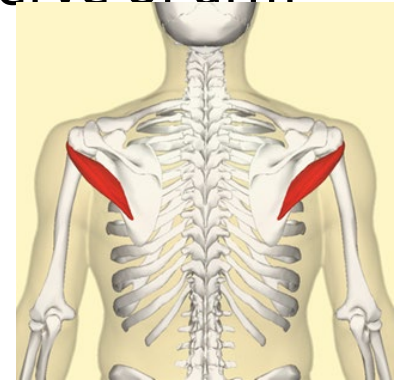


Brachial plexus branches – Posterior cord

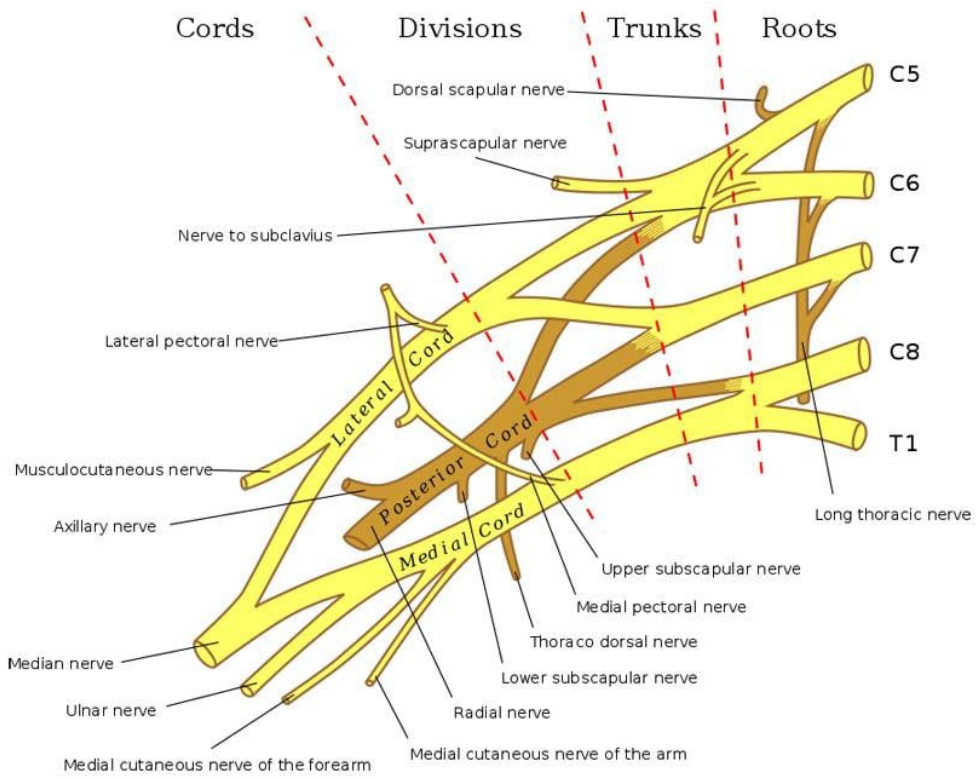


1. Upper subscapular Nerve (C5, C6)
2. Thoracodorsal nerve (C6, C7, C8)
3. Lower subscapular nerve (C5,C6)
4. Axillary nerve (C5,C6)
   - Anterior branch: deltoid and small area of overlying skin
   - Teres minor and deltoid muscle
   - Posterior branch becomes upper lateral cutaneous nerve of arm

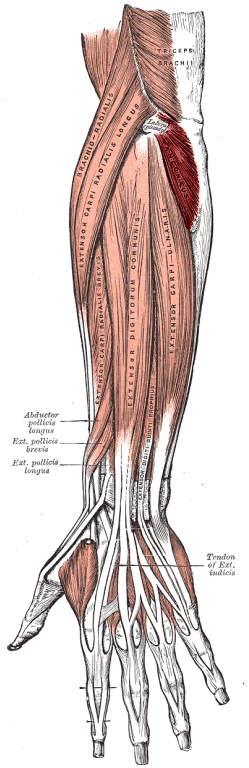




Brachial plexus branches – Posterior cord



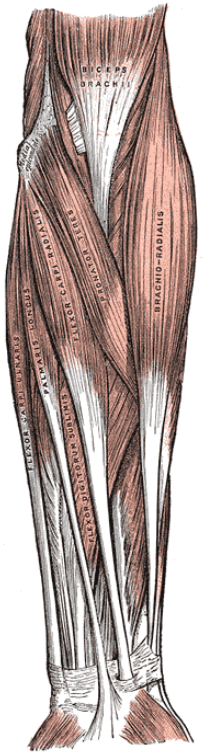
1. Upper subscapular Nerve (C5, C6)
2. Thoracodorsal nerve (C6, C7, C8)
3. Lower subscapular nerve (C5,C6)
4. Axillary nerve (C5,C6)
5. Radial Nerve (C5,C6,C7,C8 & T1)
   - All muscles of the posterior compartment arm and forearm
     - Triceps brachii, supinator, anconeus, the extensor muscles of forearm, and brachioradialis
   - Cutaneous
     - Skin of the posterior arm and forearm
     - Lower lateral surface of the arm
     - Dorsolateral surface of the hand
     - Posterior cutaneous nerve of arm (axilla)



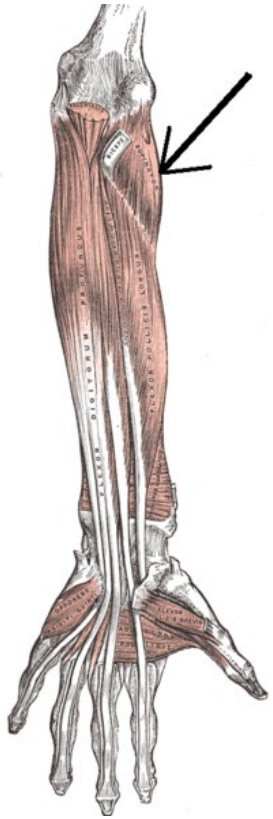
Anconeus



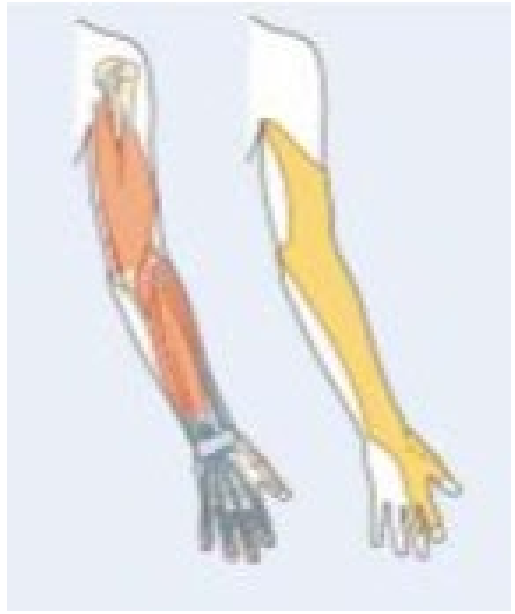
Triceps brachii



Brachioradialis



Supinator muscle



Cutaneous supply of Radial nerve

Table 7.7 Branches of brachial plexus (parentheses indicate that a spinal segment is a minor component of the nerve or is inconsistently present in the nerve)

Branch		
Dorsal scapular Origin: C5 root Spinal segment: C5		Function: motor Rhomboid major, rhomboid minor
Long thoracic Origin: C5 to C7 roots Spinal segments: C5 to C7		Function: motor Serratus anterior
Suprascapular Origin: Superior trunk Spinal segments: C5, C6		Function: motor Supraspinatus, infraspinatus
Nerve to subclavius Origin: Superior trunk Spinal segments: C5, C6		Function: motor Subclavius
Lateral pectoral Origin: Lateral cord Spinal segments: C5 to C7 (also receives contributions from spinal segments C8 to T1 through a communication with the medial pectoral nerve.)		Function: motor Pectoralis major, pectoralis minor
Musculocutaneous Origin: Lateral cord Spinal segments: C5 to C7		Function: motor All muscles in the anterior compartment of the arm Function: sensory Skin on lateral side of forearm
Medial pectoral Origin: Medial cord Spinal segments: C8, T1 (also receives contributions from spinal segments C5 to C7 through a communication with the lateral pectoral nerve)		Function: motor Pectoralis major, pectoralis minor
Medial cutaneous of arm Origin: Medial cord Spinal segments: C8, T1		Function: sensory Skin on medial side of distal one-third of arm

Table 7.7 Branches of brachial plexus (parentheses indicate that a spinal segment is a minor component of the nerve or is inconsistently present in the nerve)—cont'd

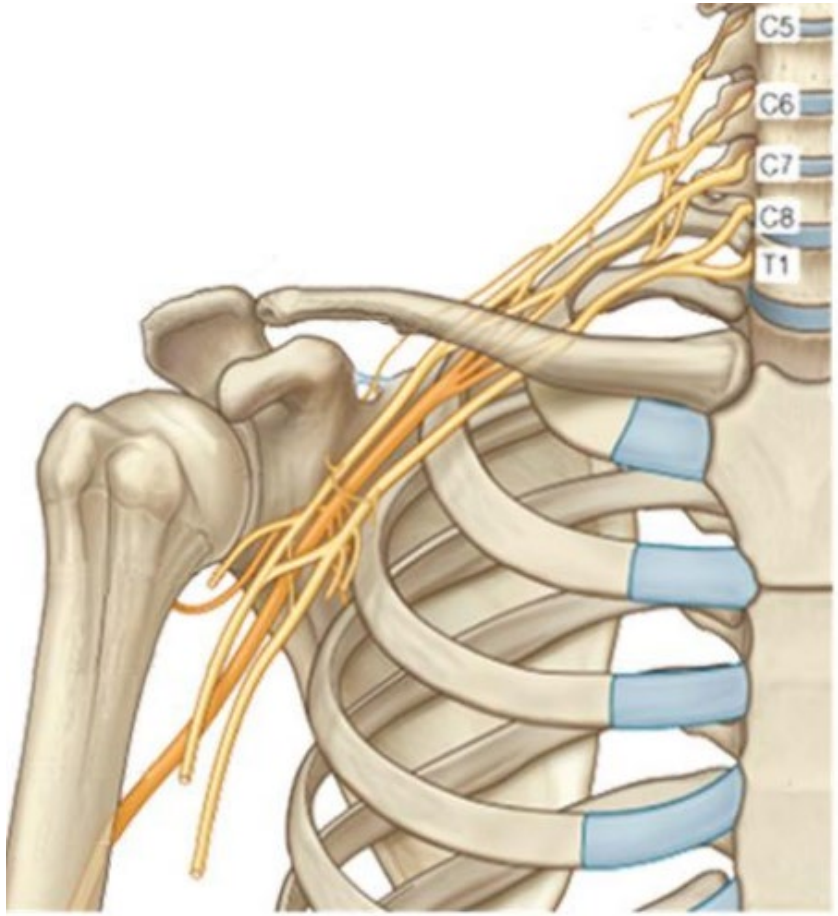
Branch		
Medial cutaneous of forearm Origin: Medial cord Spinal segments: C8, T1		Function: sensory Skin on medial side mainly of forearm
Median Origin: Medial and lateral cords Spinal segments: (C5), C6 to T1		Function: motor All muscles in the anterior compartment of the forearm (except flexor carpi ulnaris and medial half of flexor digitorum profundus), three thenar muscles of the thumb and two lateral lumbrical muscles Function: sensory Skin over the palmar surface of the lateral three and one-half digits and over the lateral side of the palm and middle of the wrist
Ulnar Origin: Medial cord Spinal segments: (C7), C8, T1		Function: motor All intrinsic muscles of the hand (except three thenar muscles and two lateral lumbricals); also flexor carpi ulnaris and the medial half of flexor digitorum profundus in the forearm Function: sensory Skin over the palmar surface of the medial one and one-half digits and associated palm and wrist, and skin over the dorsal surface of the medial one and one-half digits
Superior subscapular Origin: Posterior cord Spinal segments: C5, C6		Function: motor Subscapularis
Thoracodorsal Origin: Posterior cord Spinal segments: C6 to C8		Function: motor Latissimus dorsi
Inferior subscapular Origin: Posterior cord Spinal segments: C5, C6		Function: motor Subscapularis, teres major
Axillary Origin: Posterior cord Spinal segments: C5, C6		Function: motor Deltoid, teres minor Function: sensory Skin over upper lateral part of arm
Radial Origin: Posterior cord Spinal segments: C5 to C8, (T1)		Function: motor All muscles in the posterior compartments of arm and forearm Function: sensory Skin on the posterior aspects of the arm and forearm, the lower lateral surface of the arm, and the dorsal lateral surface of the hand

Learning Outcomes

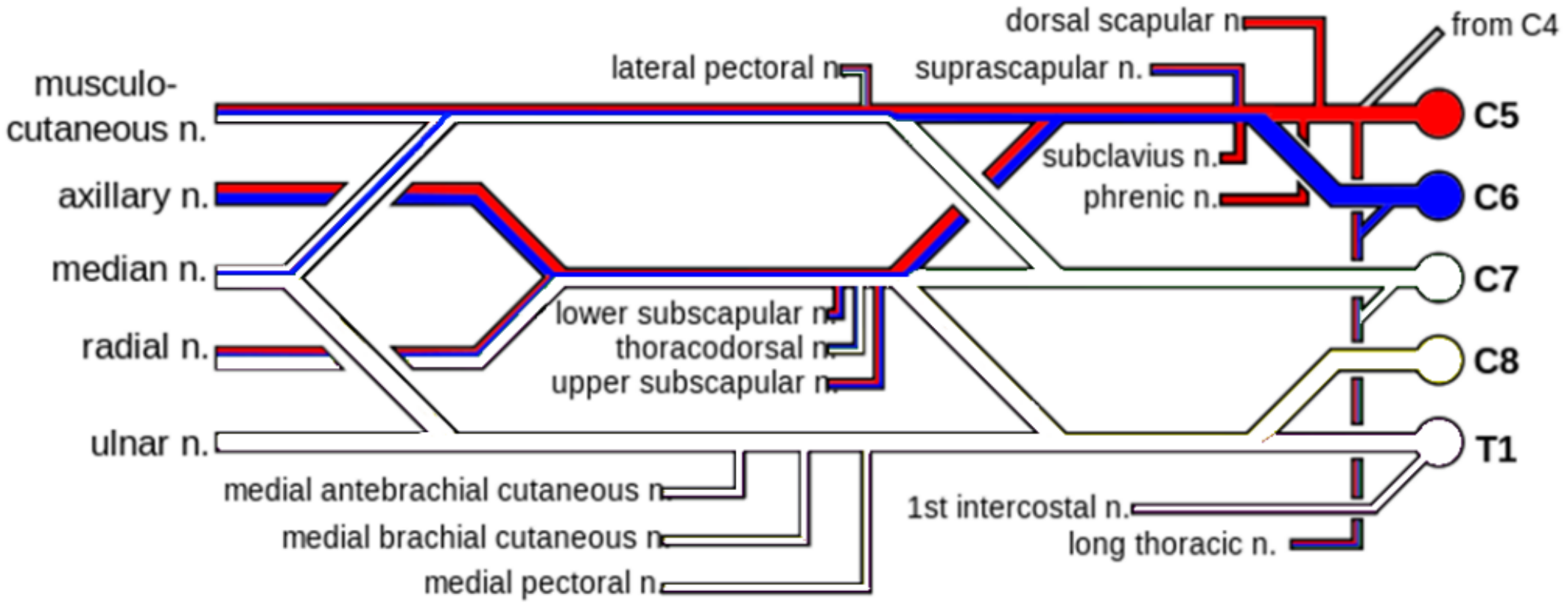
- Give an account of the distribution of cutaneous nerves of the upper limb
- Describe the pattern of dermatome innervation of the upper limb
- Describe the myotome map of the upper limb, and list the nerves responsible for movement around each joint of the upper limb
- List the origin, course and distribution of the brachial plexus
- Identify the major sites of injury to the nerves of the upper limb and outline common causes and consequences of damage to them

Injuries to the brachial plexus or terminal branches

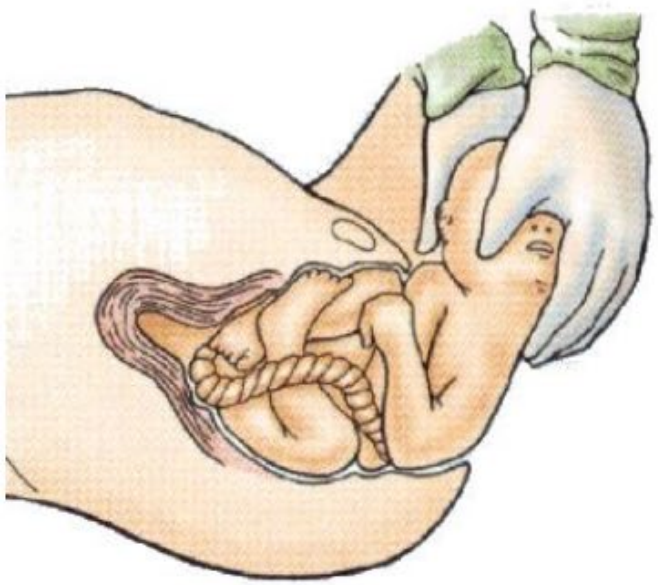
Part of the brachial plexus	Causes of injury
Roots	Compression from: • herniated cervical discs • Neck tumours • Factors responsible for thoracic outlet syndrome • Accident injuries • Obstetrics injuries
Divisions	Prone to injuries from: • clavicular fractures • Sternoclavicular dislocation
Cords	Aneurysm of the axillary artery
Terminal branches	Fractures, injuries



The Brachial Plexus Tube Map



Lesion of the upper root (C5 & C6): Erb-Duchenne's Palsy – Waiter's Tip





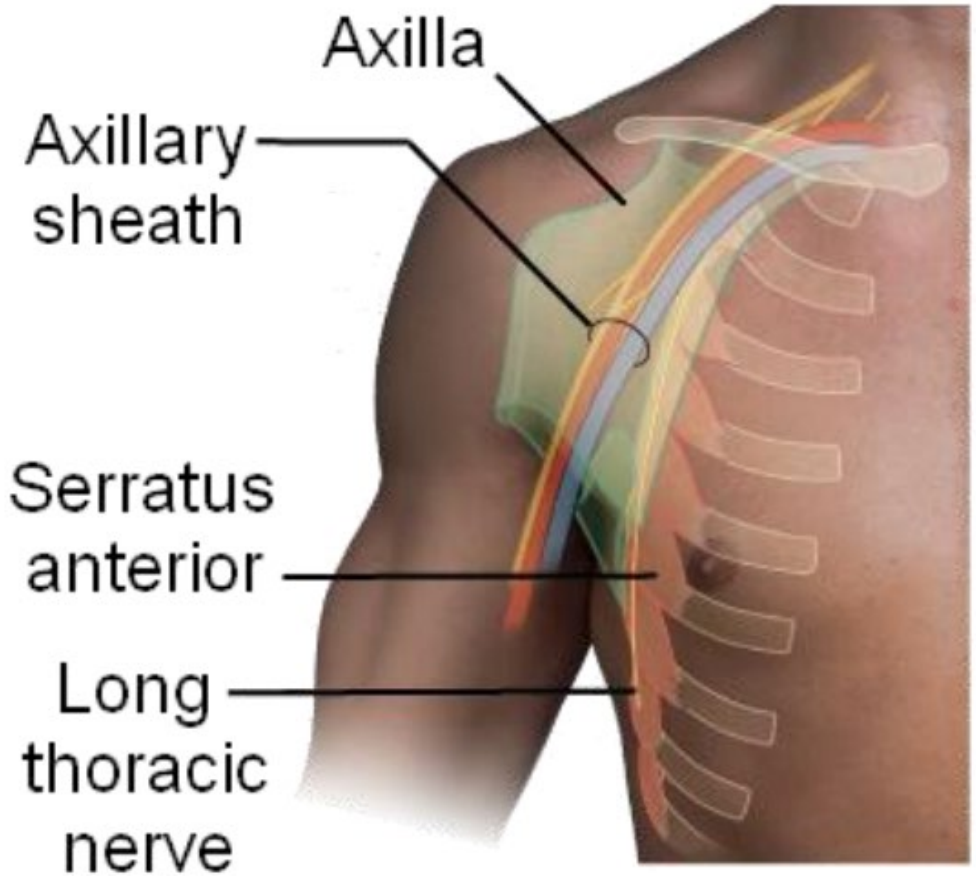


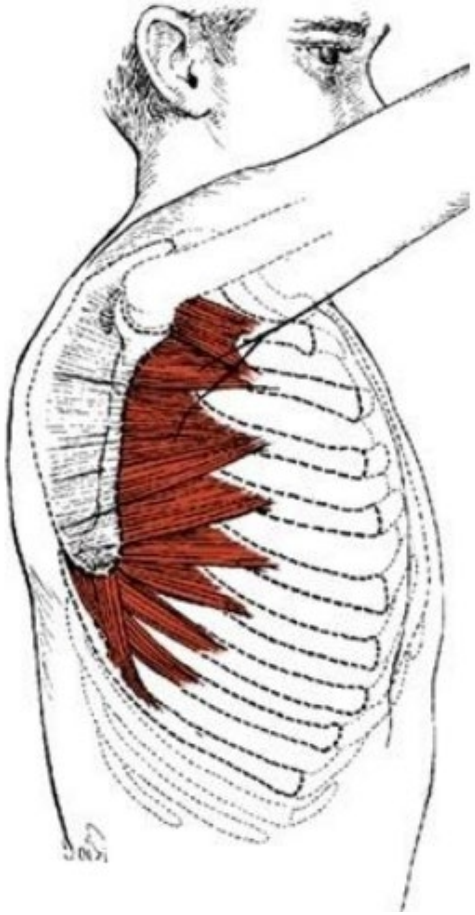
Erb-Duchenne’s Palsy Causes





Long Thoracic Nerve

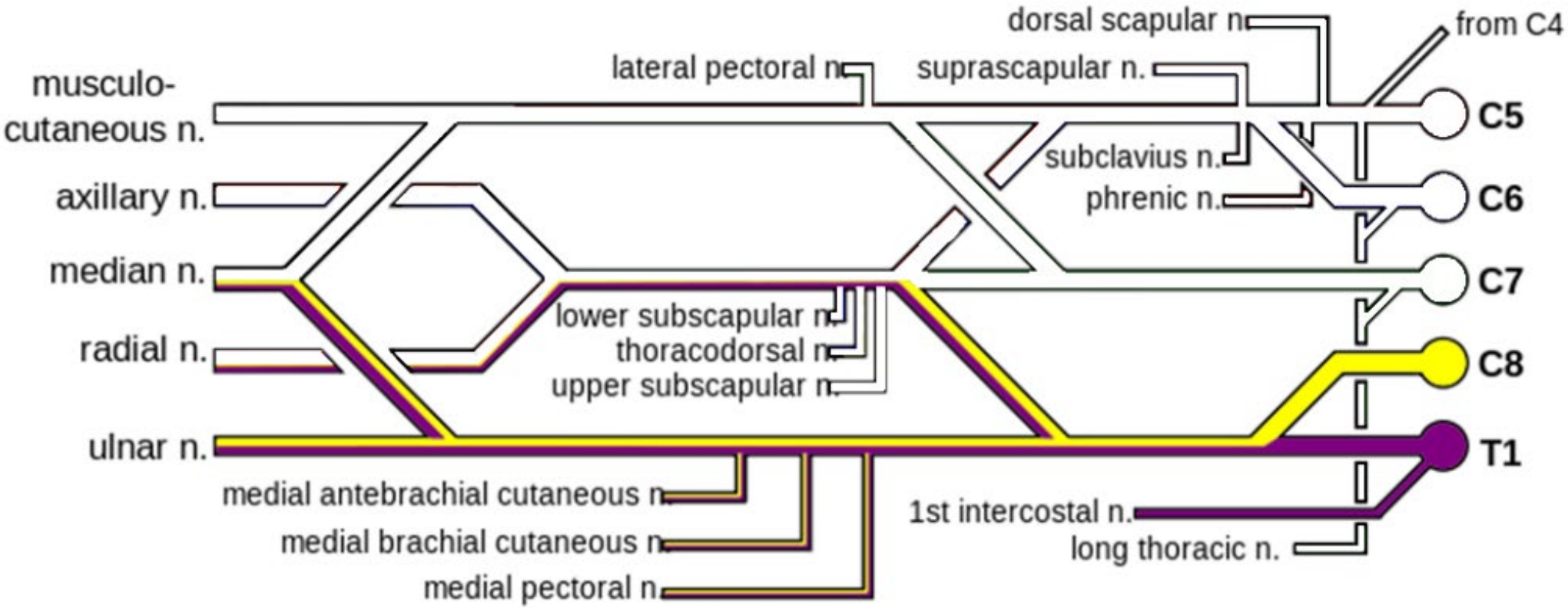




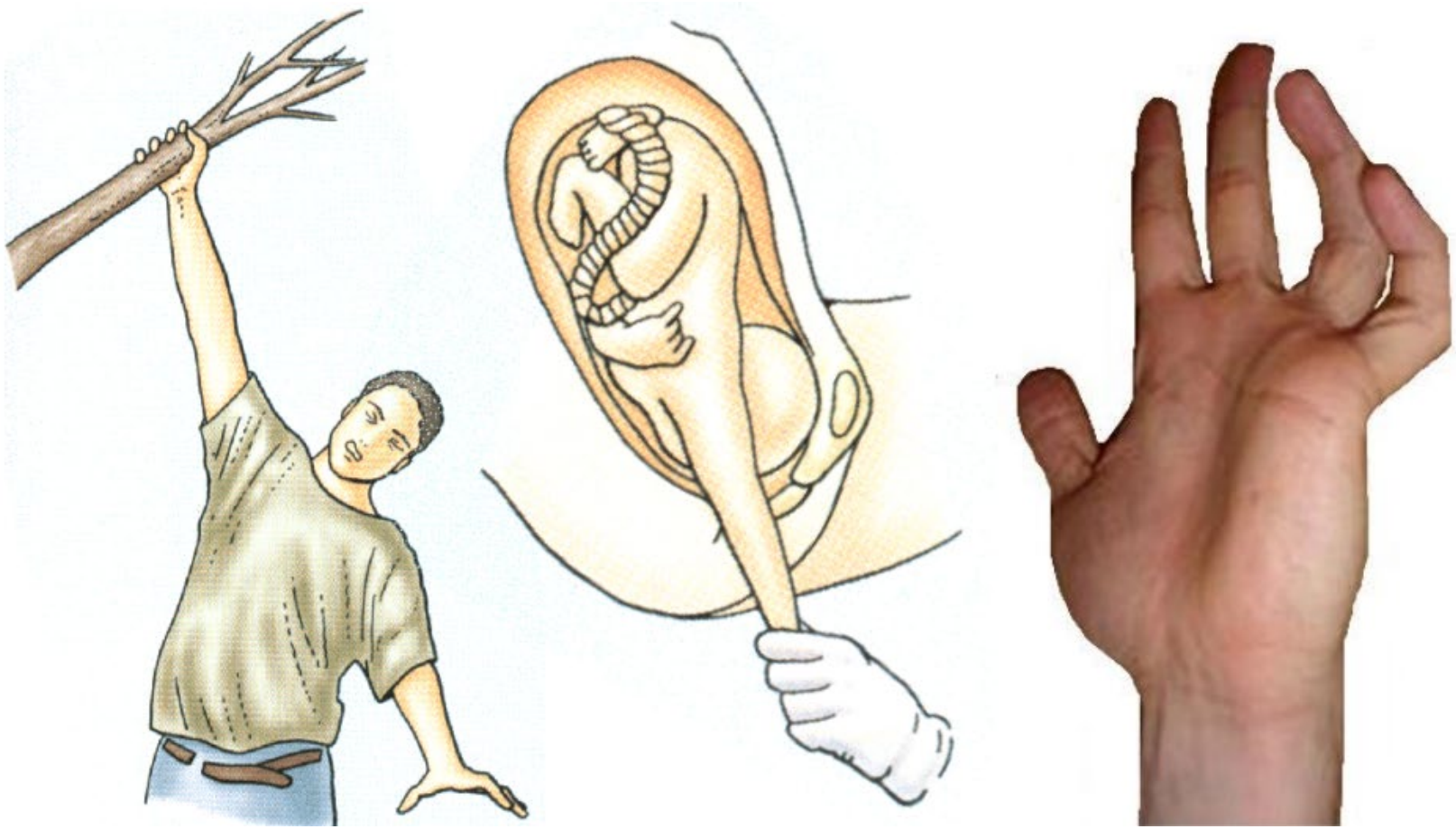


Winged scapula

Lower Roots

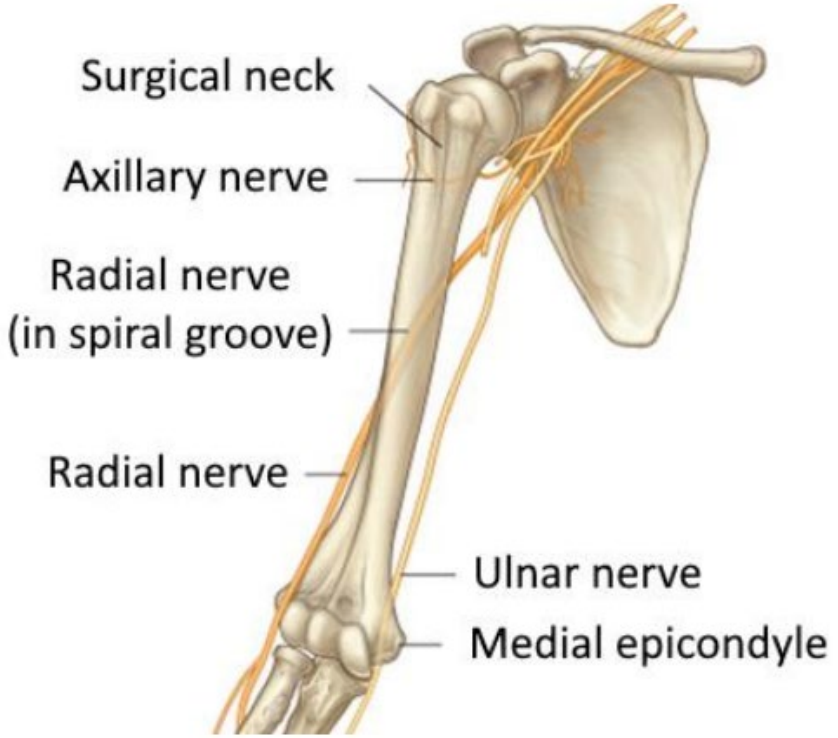

dorsal scapular n.
from C4
lateral pectoral n.
suprascapular n.
musculo-
cutaneous n.
C5
subclavius n.
axillary n.
phrenic n.
C6
median n.
C7
lower subscapular n.
radial n.
thoracodorsal n.
C8
upper subscapular n.
ulnar n.
T1
medial antebrachial cutaneous n.
1st intercostal n.
medial brachial cutaneous n.
long thoracic n.
medial pectoral n.

Injury to the lower roots (C8 & T1) - Klumpke's Palsy



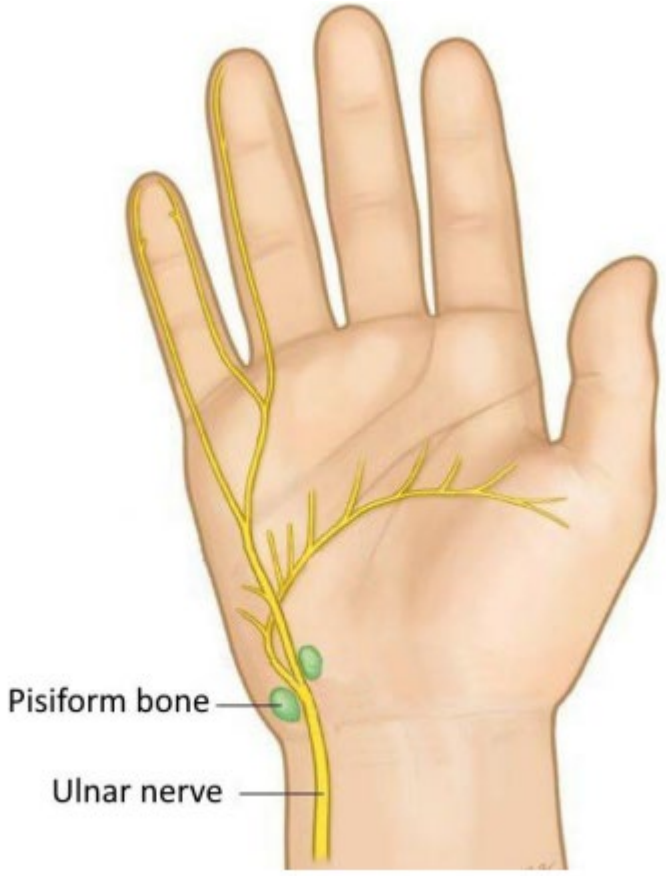
Site of Nerve Injuries

- Axillary
  - Surgical neck
- Radial
  - Spiral Groove
- Ulnar
  - Medial epicondyle



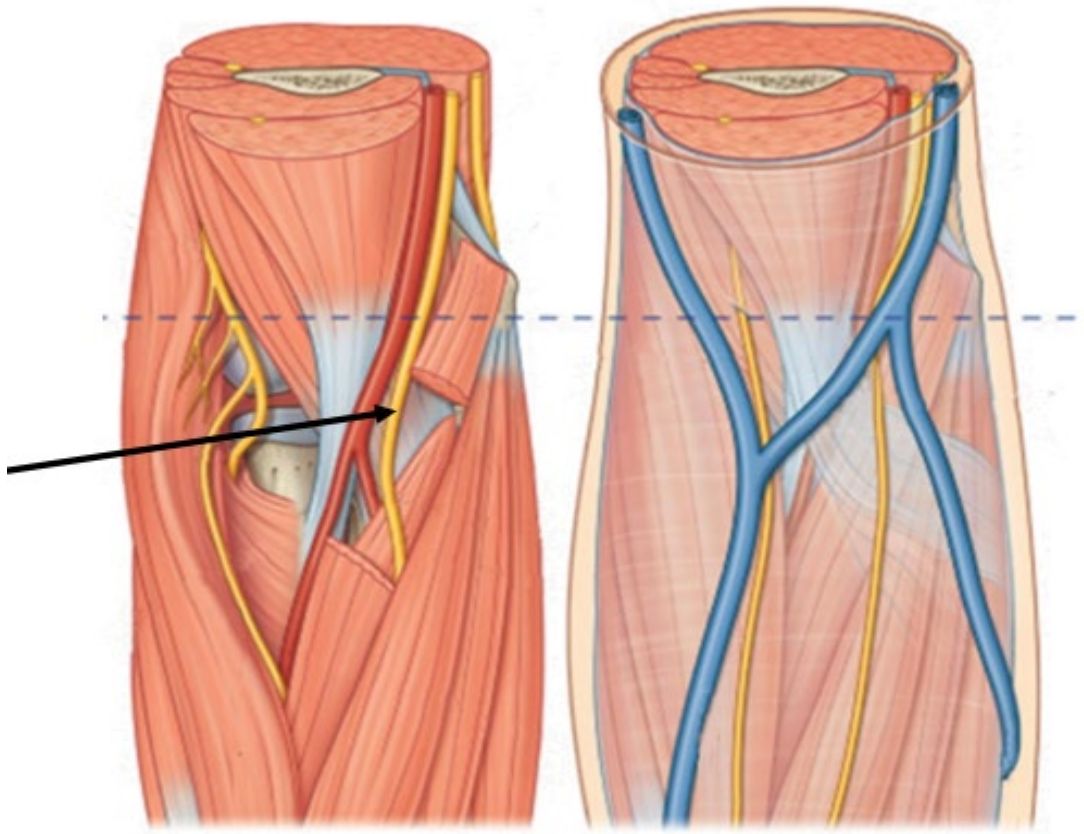
Site of Nerve Injuries

- Axillary
  - Surgical neck
- Radial
  - Spiral Groove
- Ulnar
  - Medial epicondyle
  - Pisiform
  - Guyon’s canal



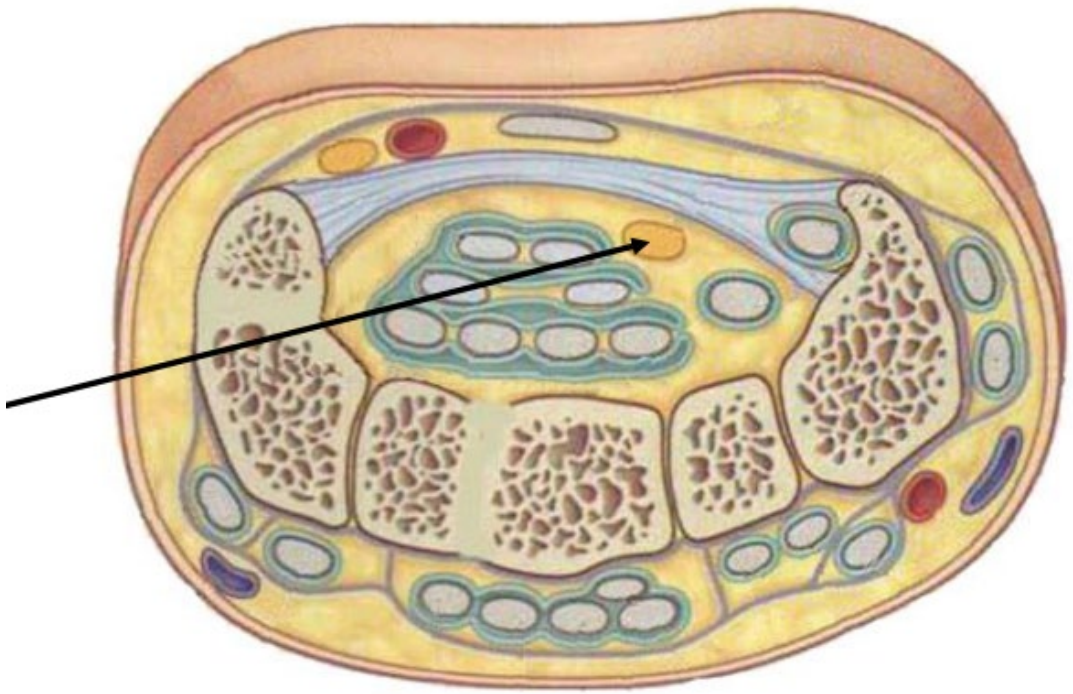
Site of Nerve Injuries

- Axillary
  - Surgical neck
- Radial
  - Spiral Groove
- Ulnar
  - Medial epicondyle
  - Pisiform
- Median
  - Cubital Fossa



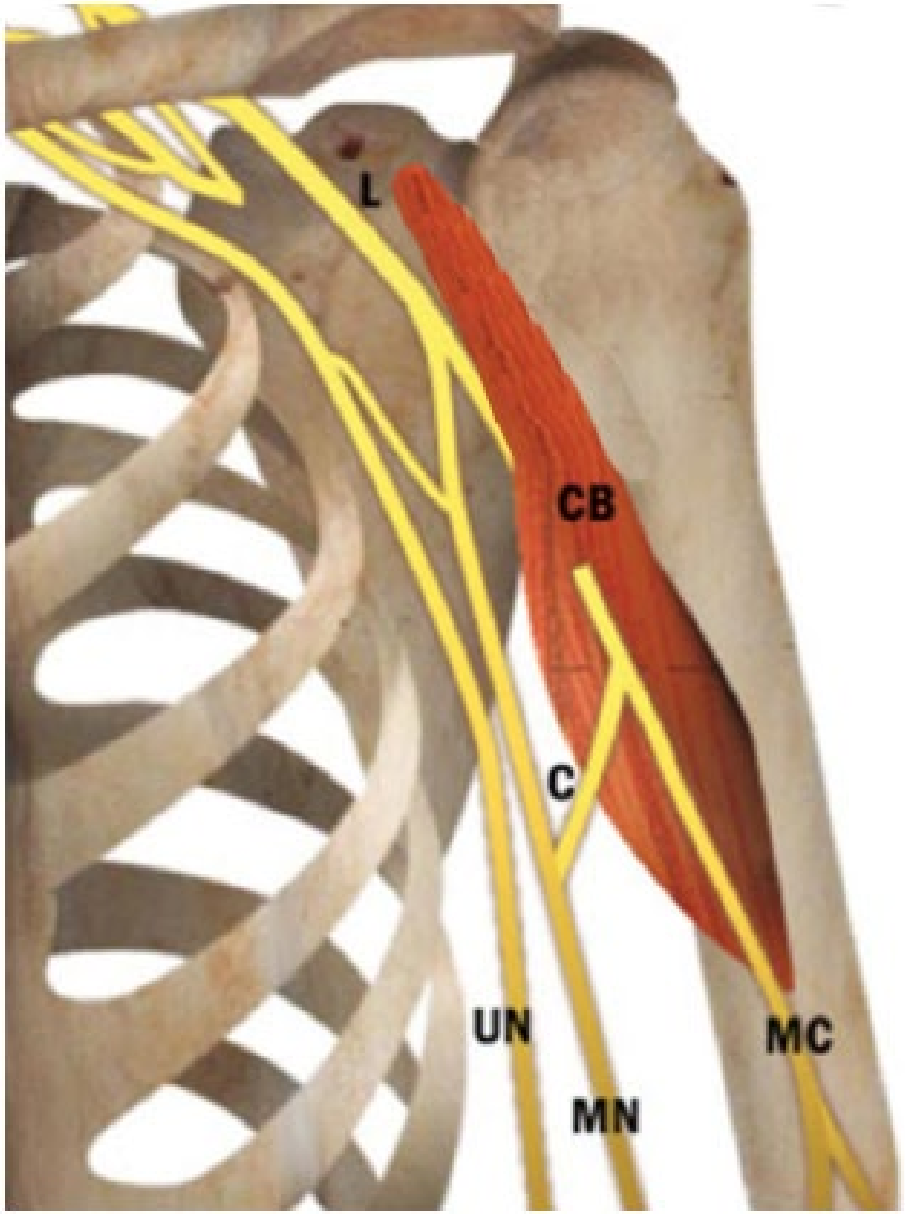
Site of Nerve Injuries

- Axillary
  - Surgical neck
- Radial
  - Spiral Groove
- Ulnar
  - Medial epicondyle
  - Pisiform
- Median
  - Cubital Fossa
  - Carpal Tunnel

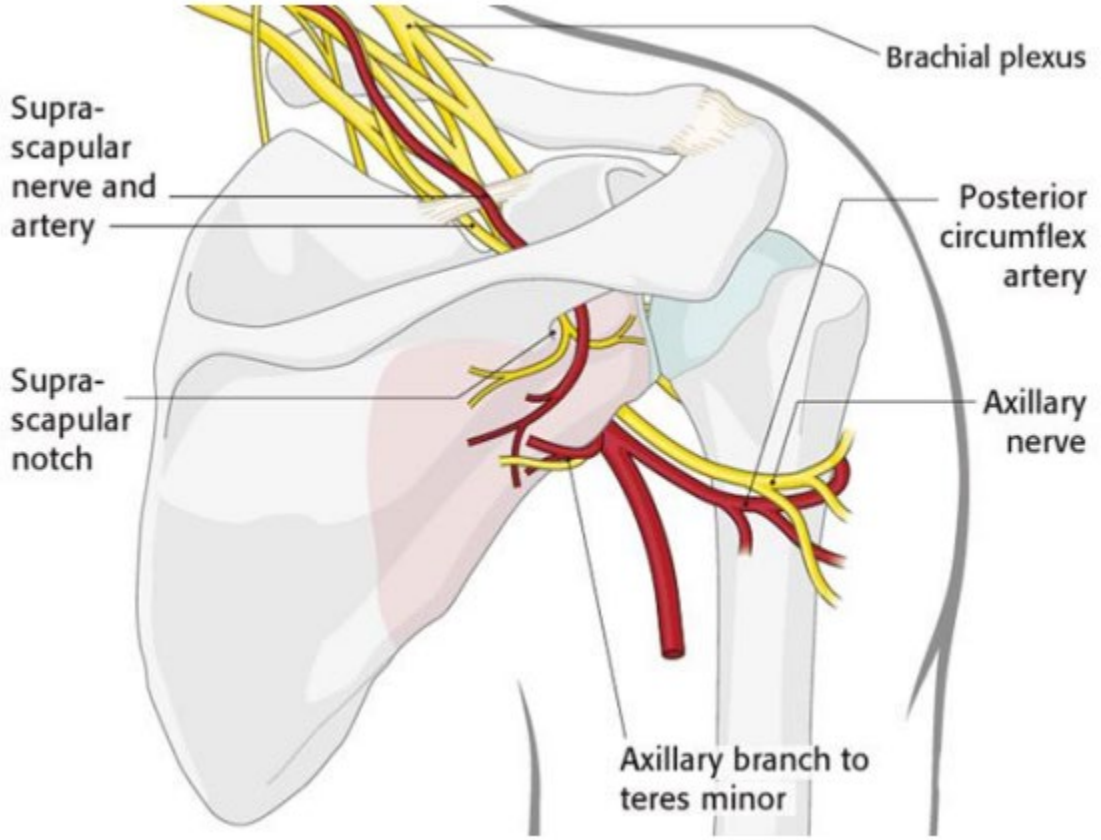


Site of Nerve Injuries

- Axillary
  - Surgical neck
- Radial
  - Spiral Groove
- Ulnar
  - Medial epicondyle
  - Pisiform
- Median
  - Cubital Fossa
  - Carpal Tunnel
- Musculocutaneous
  - Coracobrachialis

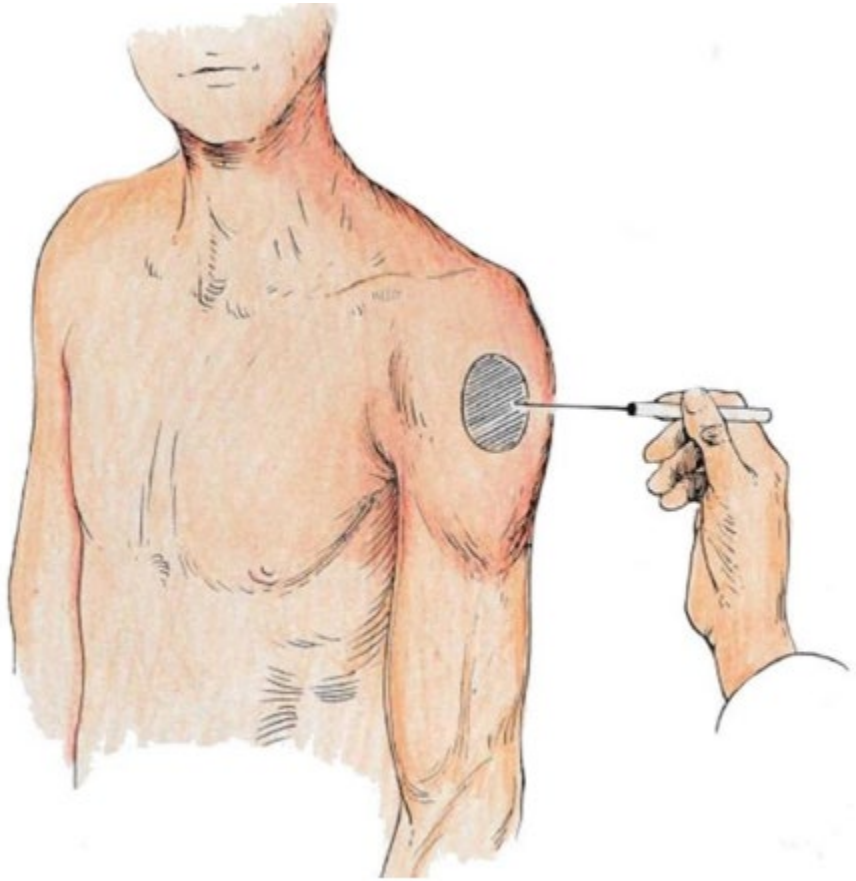


Axillary Nerve Distribution

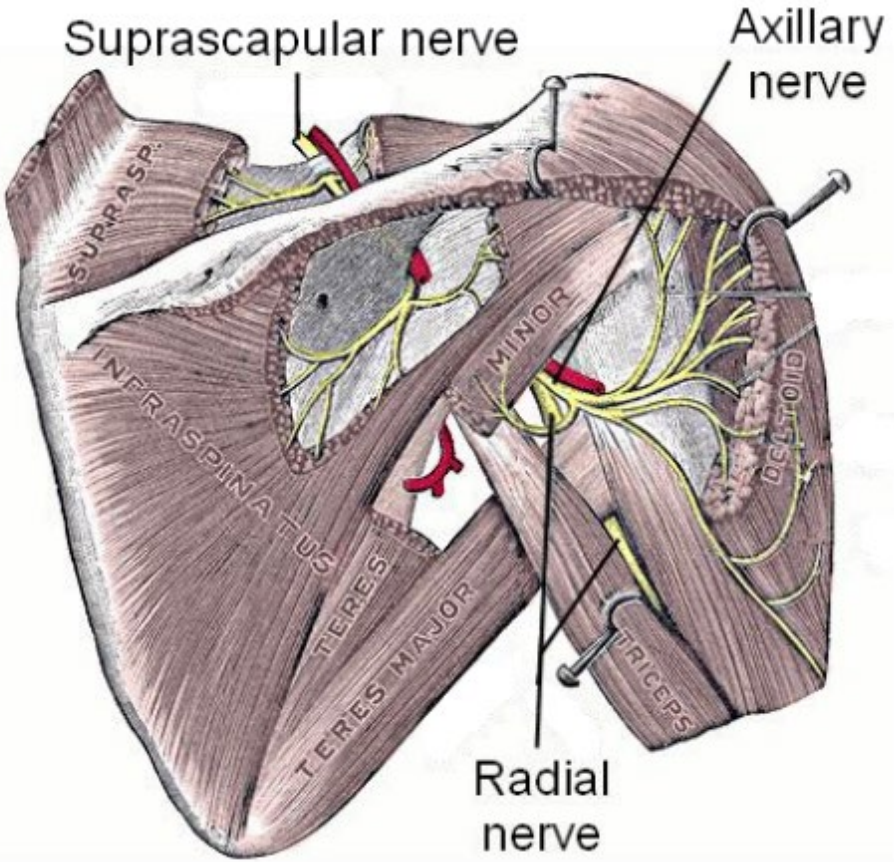


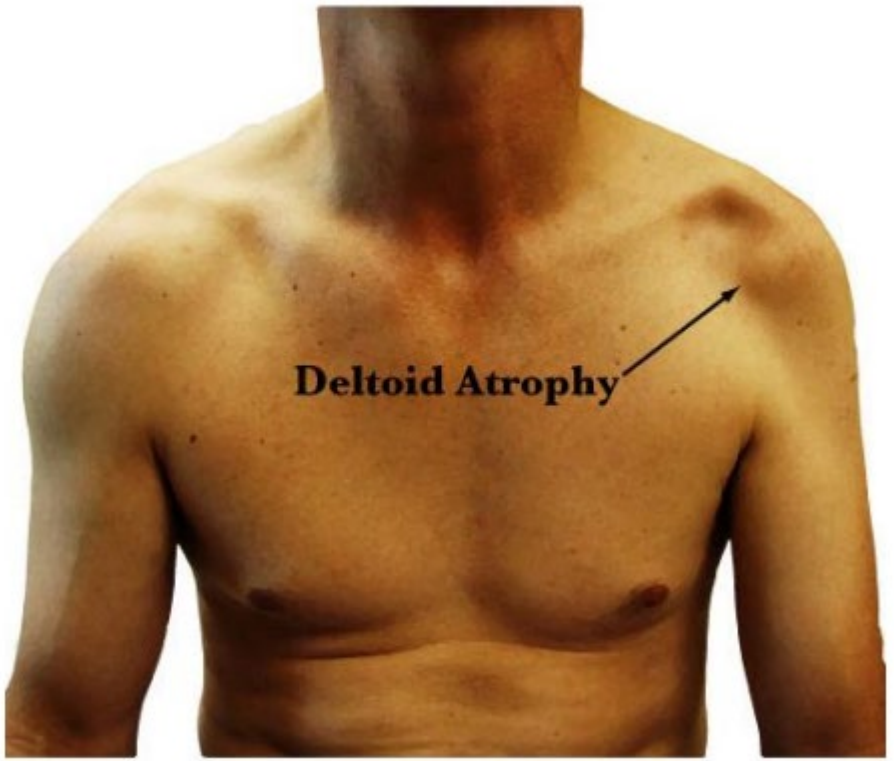
Axillary Nerve Sensory Loss





Axillary Nerve Motor Loss





Radial Nerve Distribution



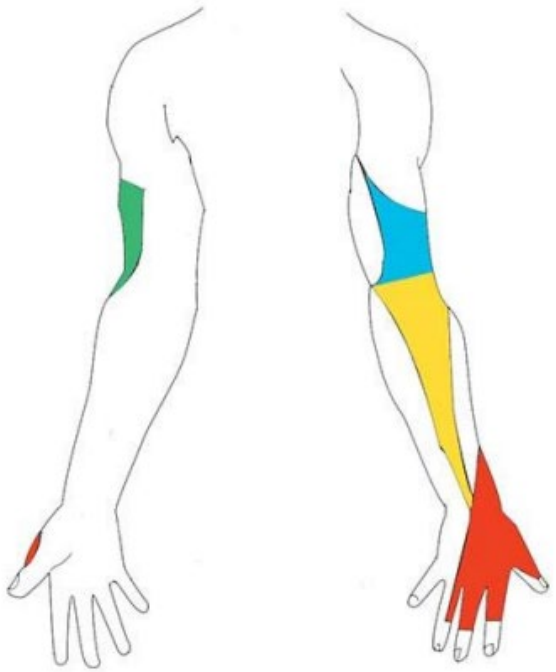
Radial Nerve Palsy causes



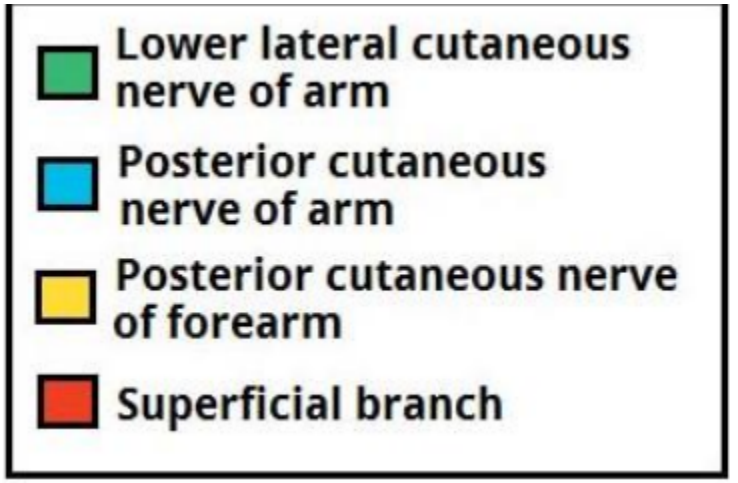




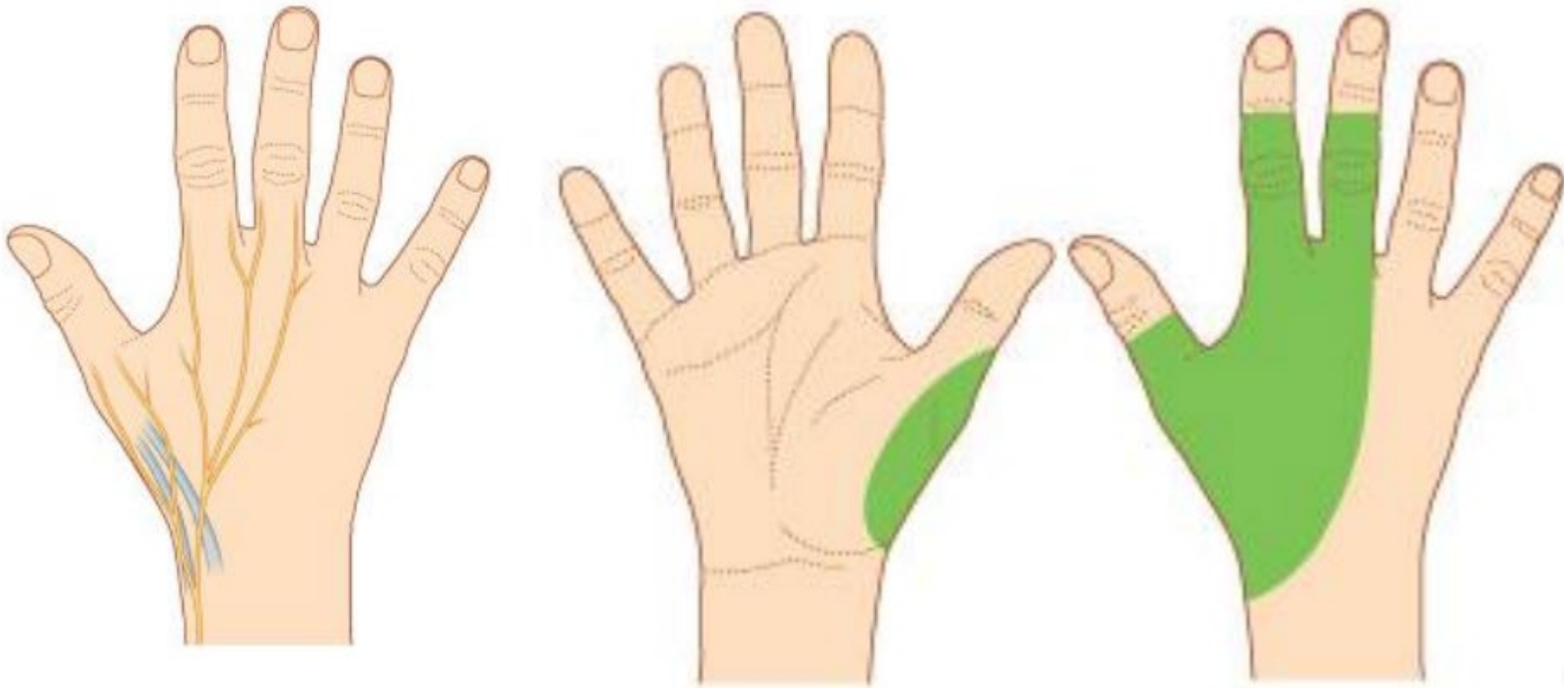
Radial Nerve Sensory Loss



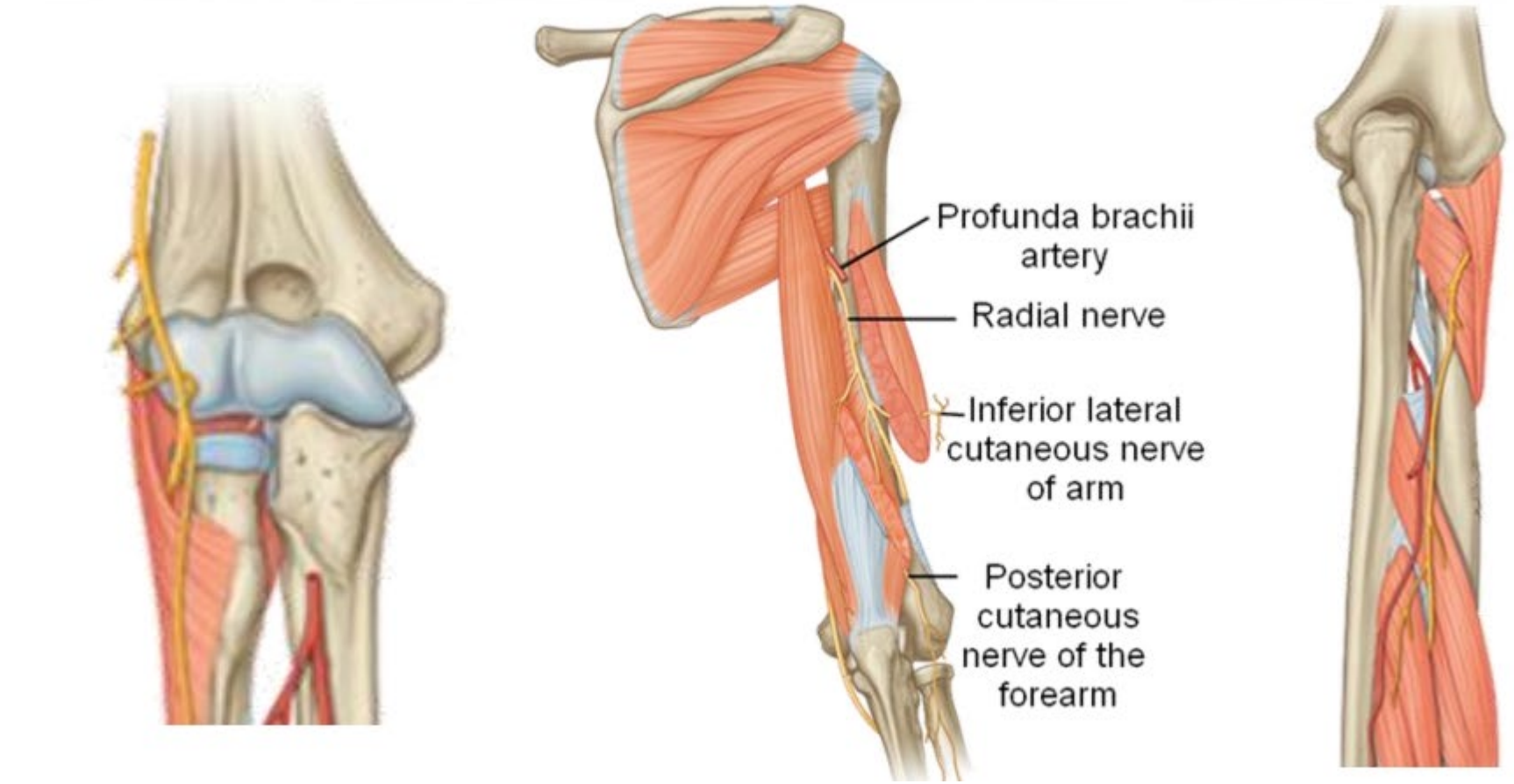




Radial Nerve Sensory Loss in Hand



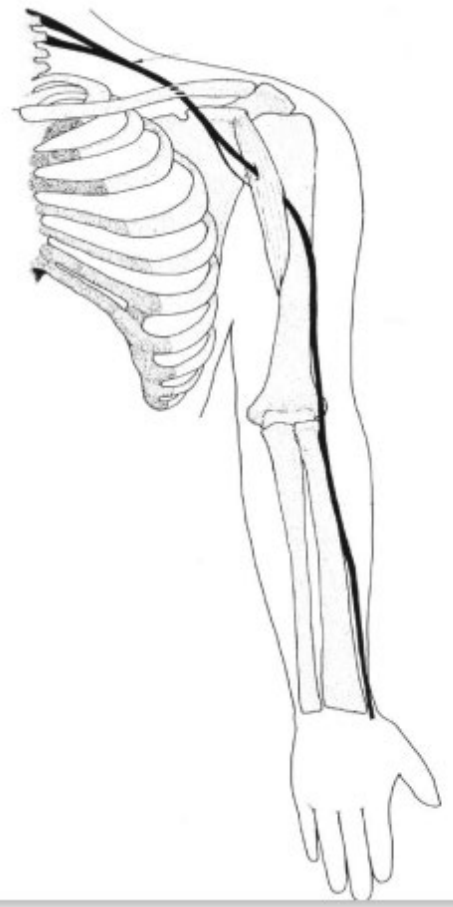
Radial Nerve Motor Loss



Radial Nerve Palsy Wrist Drop

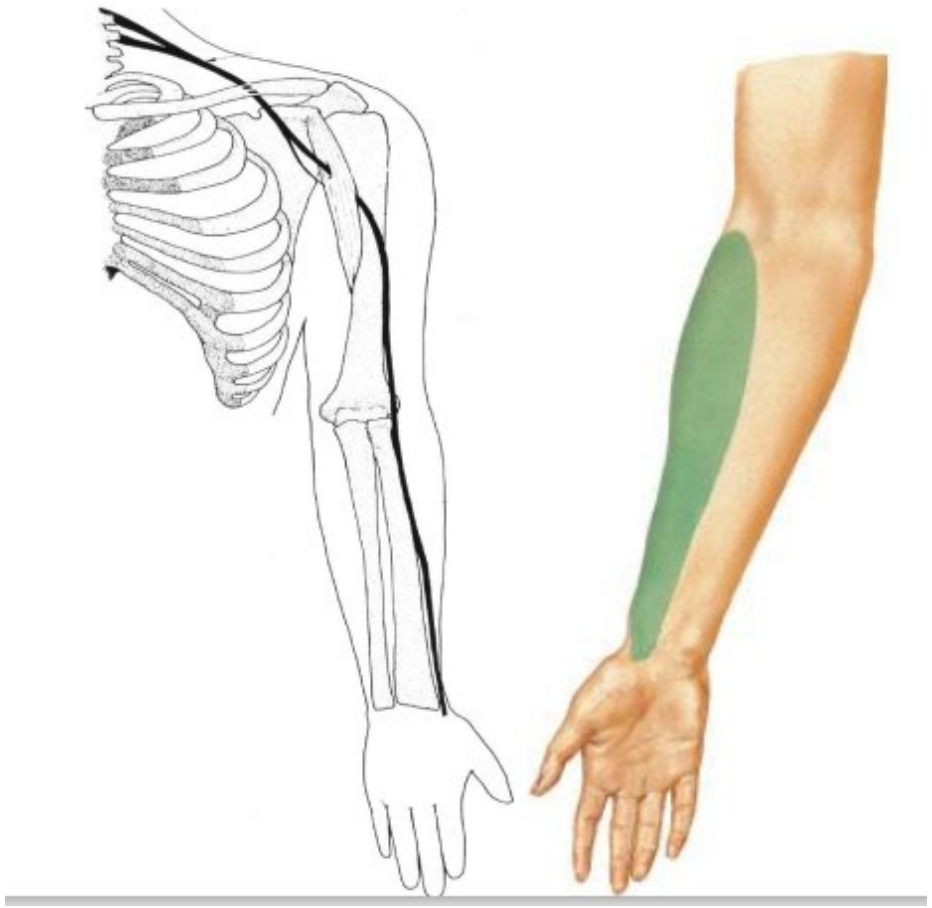


Musculocutaneous Nerve Distribution





Musculocutaneous Nerve Sensory Loss



Musculocutaneous Motor Loss

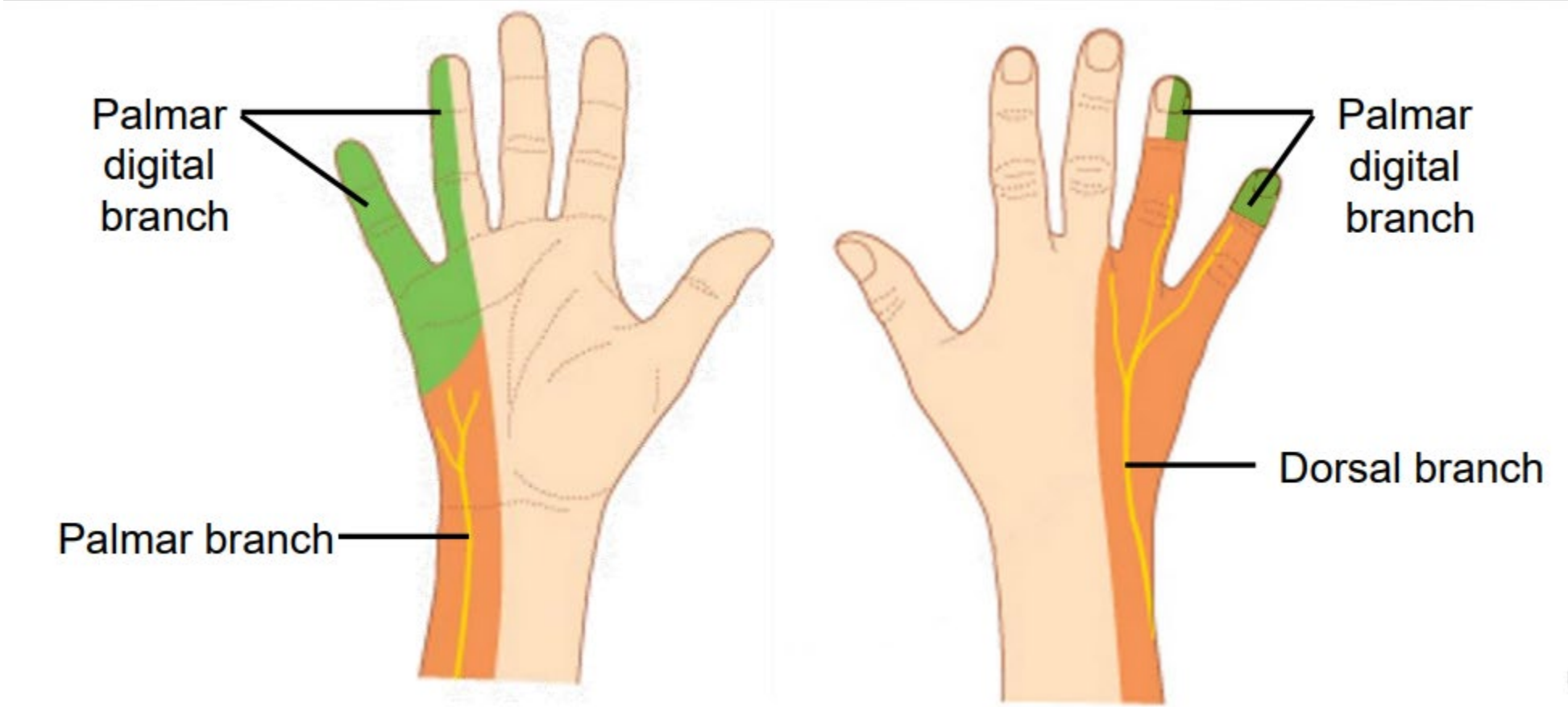




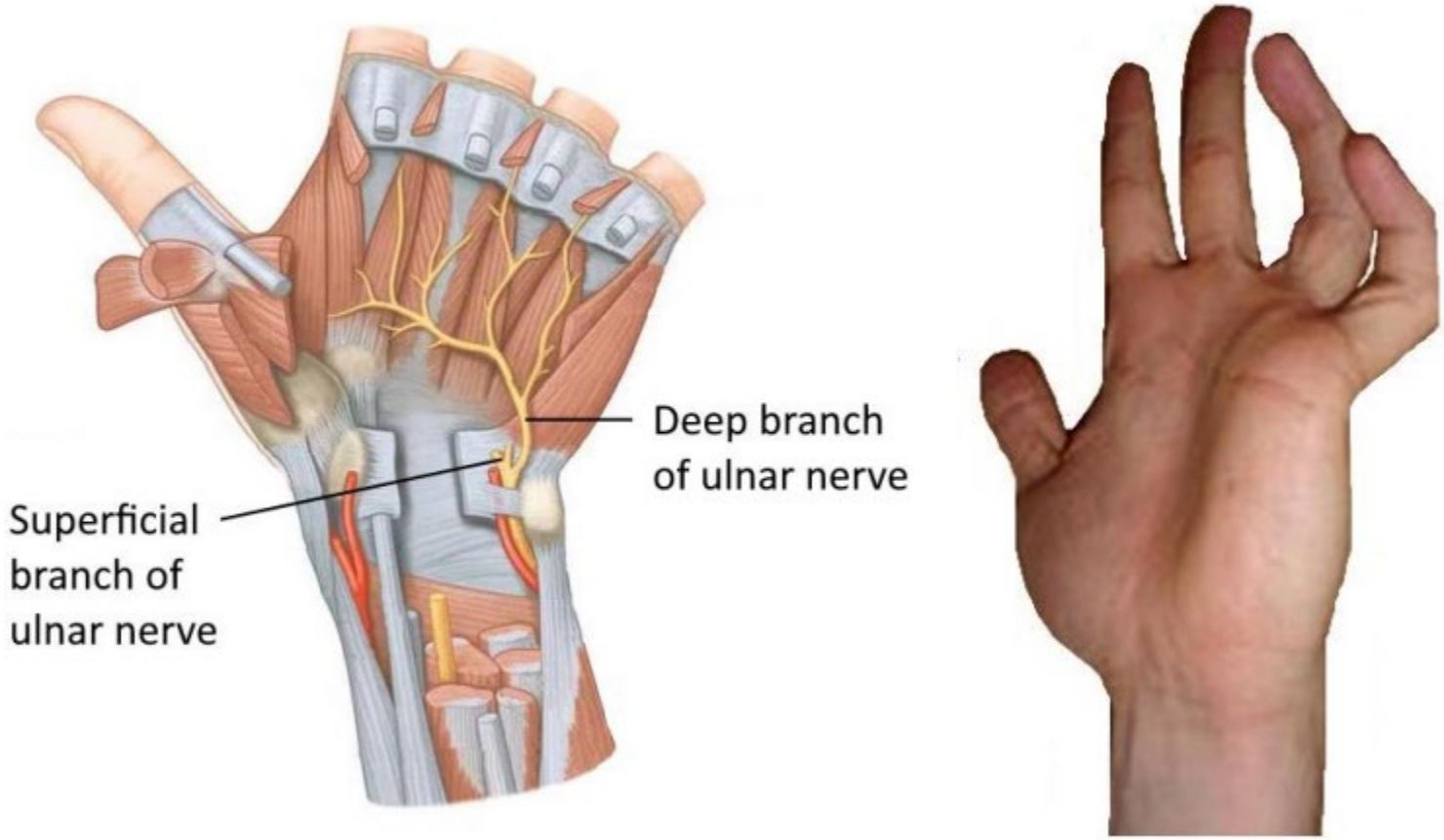
Ulnar Nerve Distribution



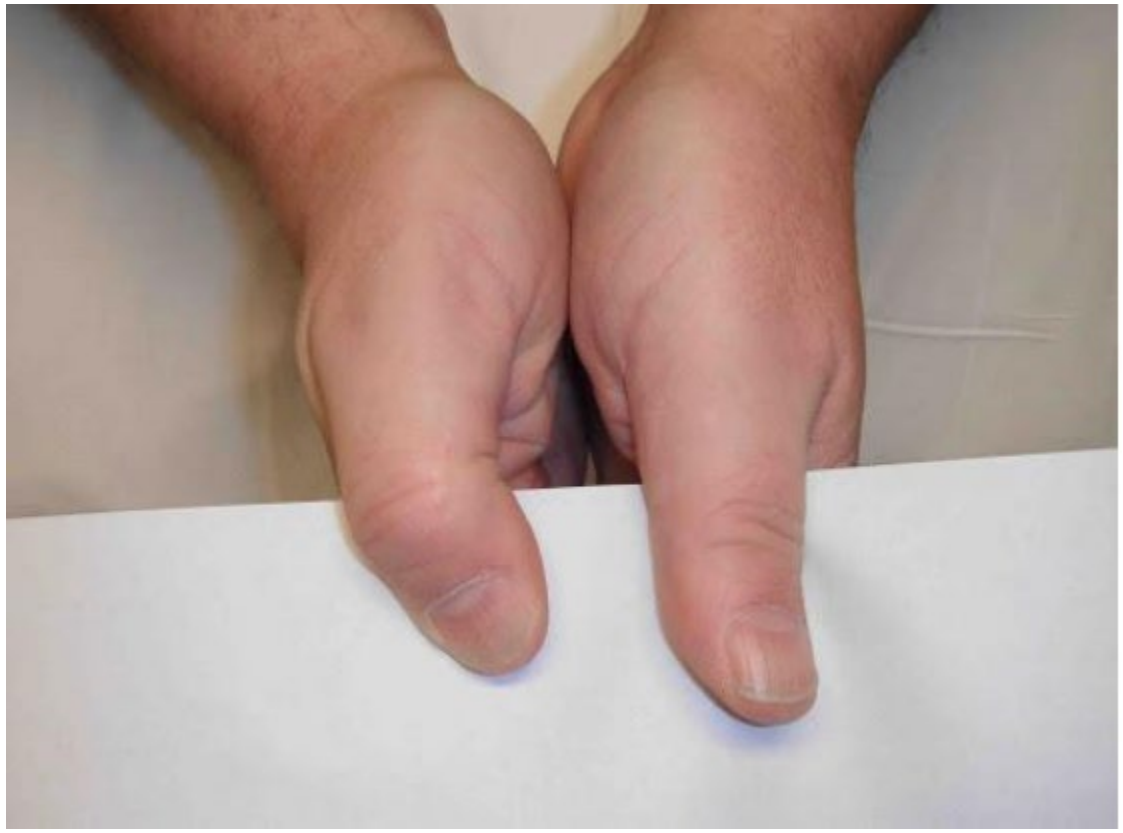
Ulnar Nerve Sensory Loss



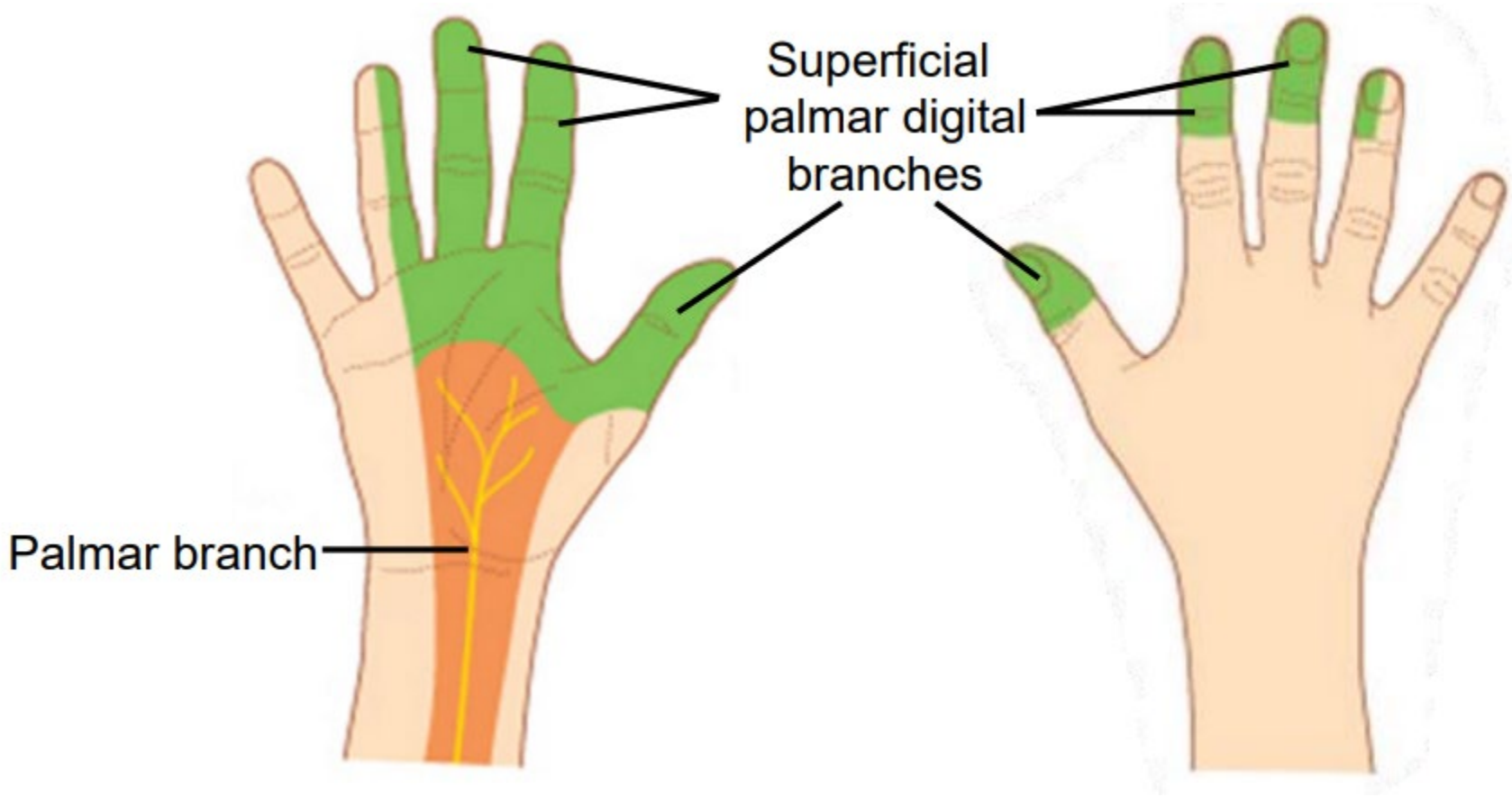
Claw Hand


Deep branch
of ulnar nerve
Superficial
branch of
ulnar nerve

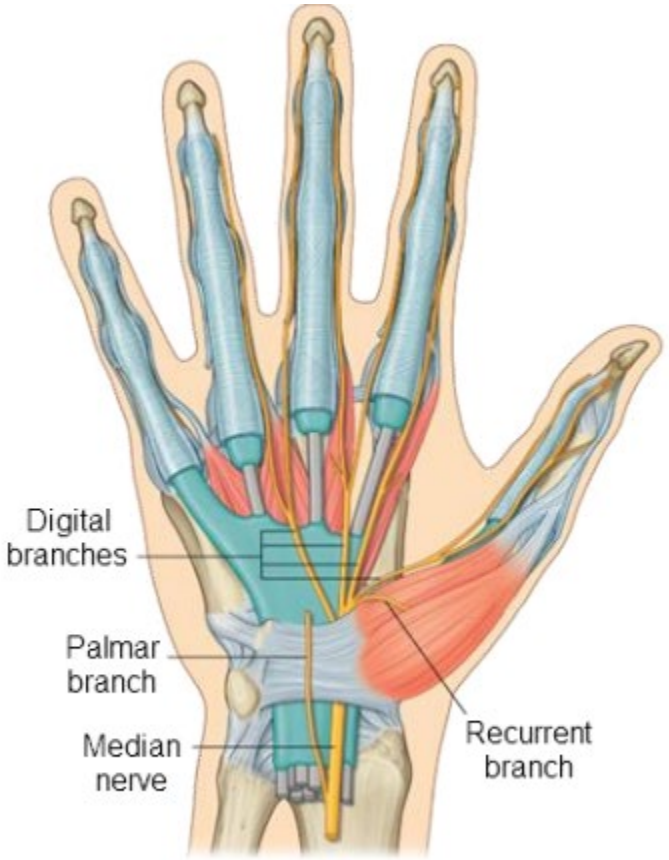
Fromet’s Sign

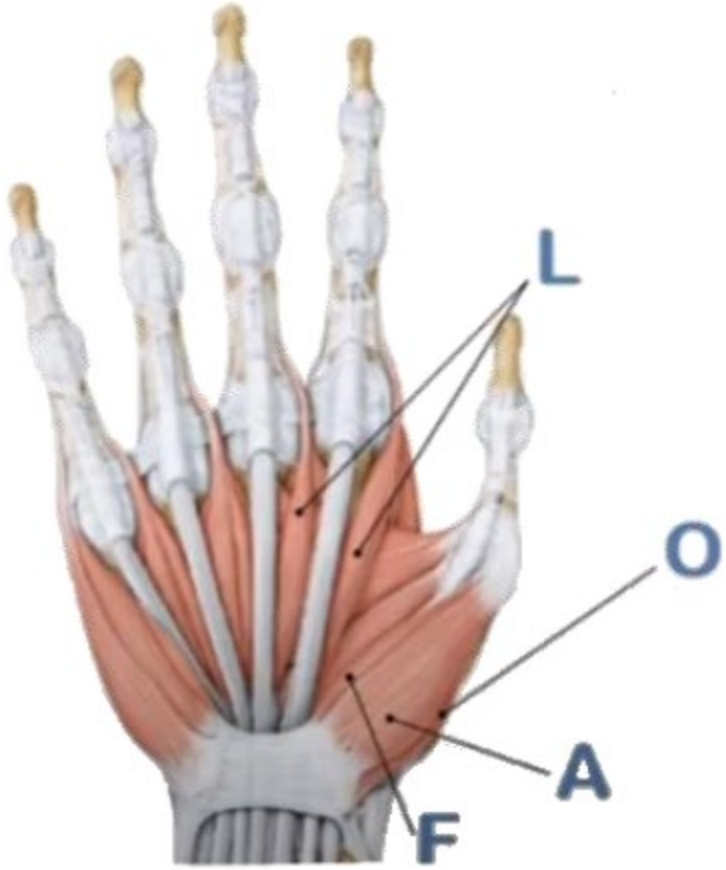


Median Nerve Sensory Loss

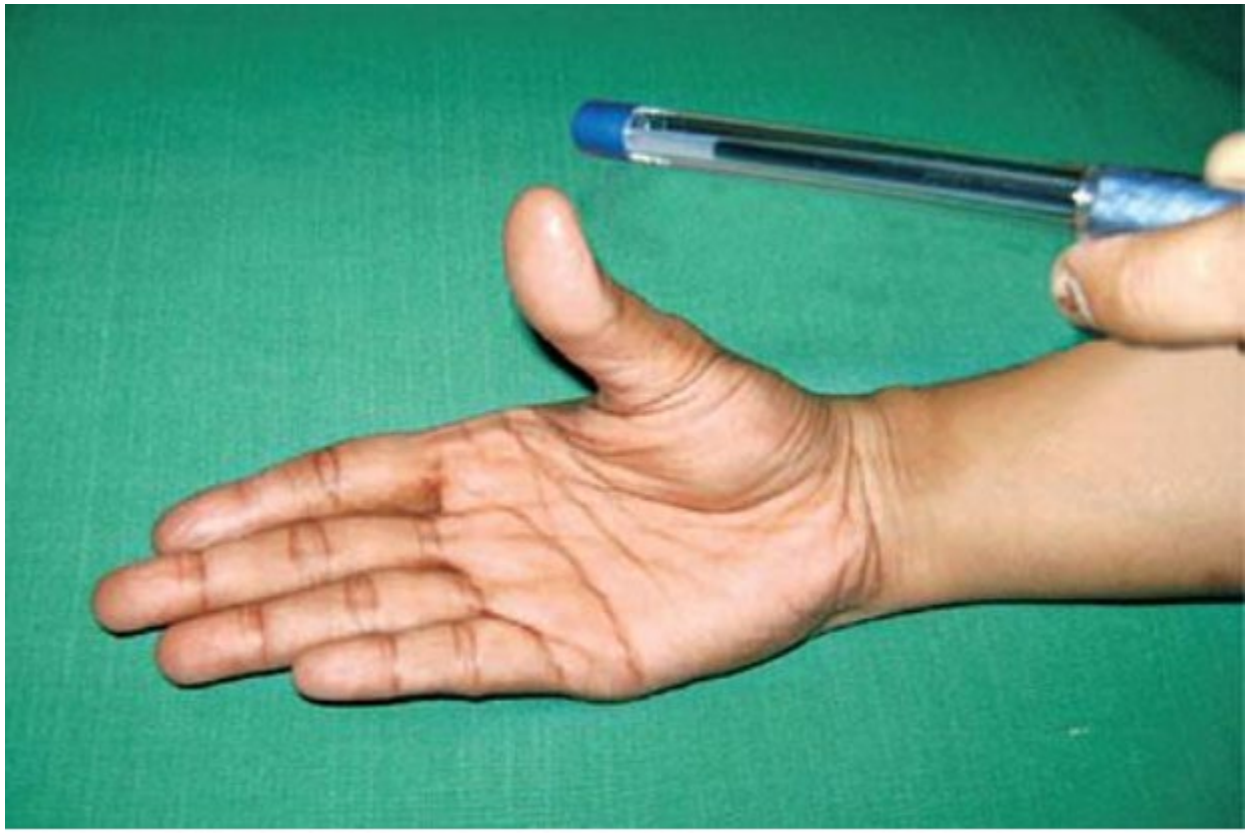


Median Nerve Motor Loss

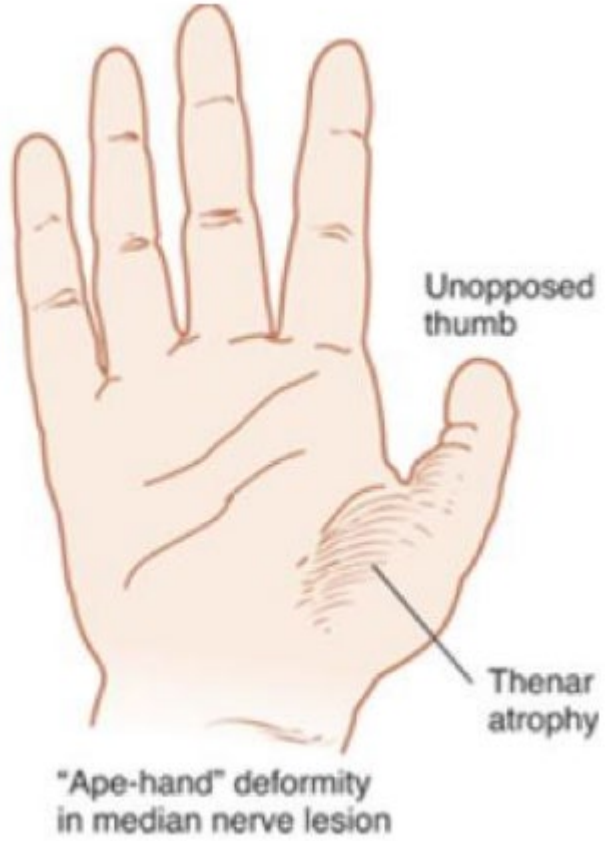




Pen Touching Test



Median Nerve Palsy Signs






Any Questions

Reference

- Drake RL, Vogl WA, Mitchell AWM. (2024). *Grays Anatomy for Students. Fifth Edition*. Philadelphia: Elservier